Munich University of Applied Sciences

Hochschule Muenchen

Department of Computer Science & Mathematics

Fakultaet fuer Informatik und Mathematik

Quality-Driven Next-Best-View Planning for Target-Conditioned Egocentric 3D Reconstruction

Qualitätsgetriebene Next-Best-View-Planung für zielkonditionierte egozentrische 3D-Rekonstruktion

Master's Thesis / Masterarbeit

for obtaining the academic degree Master of Science (M.Sc.)

submitted to
Munich University of Applied Sciences
Department of Computer Science & Mathematics

Submitted by:	Jan Duchscherer
Email:	j.duchscherer@hm.edu
Program:	Informatik - Master
Specialization:	Visual Computing and Machine Learning
Matriculation Number:	00807819

First Examiner:	Prof. Dr. Markus Friedrich
Second Examiner:	Prof. Dr. David Spieler
Supervisor:	Prof. Dr. Markus Friedrich
Submission Date:	7. Jan 2027

Transparency in the Use of AI Tools

Generative-AI tools supported literature-note organization, consistency checks, and language revision. The author selected the research questions, verified sources and technical claims, implemented and evaluated the system, and remains responsible for the submitted work.

Acknowledgements

Acknowledgements are omitted from this version.

Abstract

This thesis studies target-conditioned, quality-driven Next-Best-View (NBV) planning for egocentric 3D reconstruction in Aria Synthetic Environments (ASE). It defines target-specific Relative Reconstruction Improvement (RRI) as an oracle signal, constructs finite candidate and replay contracts, and separates privileged label generation from the actor-visible inputs available to a learned finite-horizon value model. The evaluation is designed to measure oracle-lookahead headroom and the fraction recovered by a learned policy under matched oracle re-evaluation. The evidence available for this version establishes the evaluation contract and implementation readiness, but does not contain confirmatory held-out policy outcomes; it therefore supports no claim that a learned policy improves over the specified baselines. The main remaining limitation is the absence of a validated, population-level rollout bundle with paired endpoint estimates and uncertainty.

Zusammenfassung

Diese Arbeit untersucht zielkonditionierte, qualitätsgetriebene Planung der nächsten besten Ansicht für die egozentrische 3D-Rekonstruktion in ASE. Sie definiert die zielspezifische RRI als Orakelsignal, legt Verträge für endliche Kandidatenmengen und Replay-Daten fest und trennt die privilegierte Erzeugung von Trainingssignalen von den für ein gelerntes Modell mit endlichem Horizont sichtbaren Eingaben. Die Evaluation soll den Spielraum einer vorausschauenden Orakelstrategie und den durch eine gelernte Strategie erreichten Anteil unter identischer Orakel-Neubewertung messen. Die für diese Fassung verfügbare Evidenz belegt den Evaluationsvertrag und die Implementierungsbereitschaft, enthält jedoch keine bestätigenden Ergebnisse auf zurückgehaltenen Daten. Daher wird keine Überlegenheit einer gelernten Strategie gegenüber den festgelegten Baselines behauptet. Die wesentliche verbleibende Einschränkung ist das Fehlen eines validierten Rollout-Datensatzes auf Populationsebene mit gepaarten Endpunktschätzungen und Unsicherheitsangaben.

Contents

Glossary and Abbreviations

Dataset

***ADT* – Aria Digital Twin**: Project Aria dataset with real-world captures and digital-twin scene annotations. ADT is adjacent to ARIA-NBV's sim-to-real context; the current experiments focus on ASE and EFM3D/ATEK exports, while ADT remains a relevant transfer surface.

***AEO* – Aria Everyday Objects**: Small-scale real-world Project Aria object dataset used by EFM3D for egocentric 3D perception evaluation. AEO is relevant as a possible sim-to-real check for ARIA-NBV ideas that are first developed on ASE mesh-supervised snippets.

***ASE* – Aria Synthetic Environments**: Large-scale synthetic indoor dataset with simulated Project Aria sensor characteristics, egocentric trajectories, and scene annotations. ARIA-NBV uses ASE snippets and the public mesh-supervised subset to generate oracle RRI labels from ground-truth meshes and semi-dense point clouds.

***CPF* – Central Pupil Frame**: Coordinate frame placed at the midpoint between the left and right eye boxes of Project Aria glasses. CPF is used by Project Aria for gaze-related quantities and should remain distinct from rig, camera, world, and PyTorch3D frames in ARIA-NBV docs.

***MPS* – Machine Perception Services**: Project Aria processing services that derive pose, mapping, gaze, hand, and related perception outputs from sensor recordings. ARIA-NBV mainly depends on MPS-style pose and semi-dense mapping products as the current reconstruction state for RRI computation.

***MTD* – Motion Trajectory Data**: Device poses over time, usually represented as a sequence of 6-DoF transformations. MTD supplies the logged egocentric trajectory used to define snippet state, current reconstruction context, and candidate-view reference poses.

***MFCD* – Multi-Frame Camera Data**: Synchronized camera streams from multiple Project Aria cameras over a temporal window. MFCD provides the egocentric image evidence consumed by EVL and aligned with pose, calibration, and semi-dense point streams.

***MSDPD* – Multi-Semi-Dense Point Data**: Semi-dense 3D point observations generated by SLAM-style processing across a snippet or trajectory window. MSDPD is the sparse observed geometry that ARIA-NBV fuses with candidate point clouds and compares against ground-truth meshes for RRI labels.

***Project Aria* – Project Aria**: Meta's egocentric research-device and tooling ecosystem for calibrated, time-aligned multimodal sensing. ARIA-NBV treats Project Aria and its MPS-style products as the actor-visible sensing contract, while ASE meshes remain offline supervision and evaluation assets.

***SSL* – SceneScript Language**: Structured language representation for indoor scene layout using primitives such as walls, doors, windows, and objects. SceneScript is relevant to ARIA-NBV as a possible semantic/global planning layer and as one source of ASE scene-structure context.

***snippet* – Snippet**: Short synchronized temporal window of Aria sensor data used as one EVL/VIN input sample. A snippet typically contains RGB or grayscale streams, poses, calibration, semi-dense points, and scene metadata that EVL lifts into a voxel grid.

***VRS* – Virtual Reality Standard**: File format used by Project Aria tooling to store multi-modal sensor recordings efficiently. VRS is part of the upstream Project Aria ecosystem, while ARIA-NBV primarily consumes ASE/ATEK-derived tensorized snippets for current experiments.

Entity

***target* – Target of Interest**: Selected entity, object crop, point, region, or surface-deficit hypothesis whose reconstruction quality should be improved. Target-conditioned ARIA-NBV variants use this target as an explicit input to candidate scoring and planning instead of optimizing only scene-level RRI.

Geometry

***DoF* – Degrees of Freedom**: Number of independent pose parameters available to a camera, object, or action representation. ARIA-NBV distinguishes full 6-DoF pose estimation from reduced 5-DoF candidate or action spaces used for practical view planning.

***frustum* – Frustum**: Truncated pyramidal camera-visible volume bounded by near and far clipping planes plus lateral field-of-view planes. Candidate-view frusta define which scene surfaces can project into a camera image and are therefore central to visibility, rendering, and RRI diagnostics.

***LUF* – Left-Up-Forward**: Camera coordinate convention whose x axis points left, y axis points up, and z axis points forward. LUF is one of the coordinate-frame conventions that must stay explicit when moving between Project Aria cameras, PyTorch3D cameras, and ARIA-NBV candidate poses.

***OBB* – Oriented Bounding Box**: 3D bounding box with arbitrary orientation, used to represent object extent more tightly than an axis-aligned box. OBBs are a natural target encoding for entity-aware ARIA-NBV because they provide center, extent, orientation, semantic class, and confidence signals.

***6DoF* – Six Degrees of Freedom**: Pose parameterization with three translational and three rotational degrees of freedom. Project Aria poses, candidate cameras, and world-to-device transforms are usually represented as 6-DoF rigid-body transforms.

Metrics

***AUC* – Area Under Curve**: Aggregate score computed by integrating a metric curve over an acquisition, threshold, or ranking axis. AUC can summarize how quickly reconstruction quality, coverage, or ranking quality improves as views are acquired or candidate thresholds change.

***CD* – Chamfer Distance**: Historical bidirectional distance family used to compare reconstructed points against reference geometry. Thesis-facing ARIA-

NBV notation uses point-mesh error D with directional components $D_{P \to M}$ and $D_{M \to P}$; older seminar material may still call this CD.

***CR* – Coverage Ratio**: Fraction of a target surface, scene, or region treated as observed under a chosen visibility or distance threshold. Coverage ratio is a useful diagnostic baseline for NBV, but ARIA-NBV treats reconstruction-quality improvement through RRI as the preferred optimization target.

***GT* – Ground Truth**: Reference data or annotations treated as the trusted target for training, validation, or evaluation. In ARIA-NBV, ground truth usually refers to ASE meshes, object annotations, poses, or labels used to compute oracle supervision and diagnostic metrics.

***RRI* – Relative Reconstruction Improvement**: Metric quantifying the relative reconstruction-quality improvement obtained by adding a candidate observation to the current reconstruction. ARIA-NBV computes RRI by comparing reconstruction error before and after fusing candidate-view geometry, usually against an ASE ground-truth mesh for oracle supervision and evaluation.

***reward* – Target Root-Gain Reward**: Quality-only immediate reward for thesis-core target-aware rollouts. It is the target-specific point-mesh error reduction normalized by the rollout-root target error; scalar motion, rule, diversity, and validity penalties are later ablations.

***target RRI* – Target-Specific RRI**: RRI computed only on the ground-truth and reconstructed geometry associated with a selected target of interest. Target-specific RRI lets the thesis compare views by how much they improve a selected object or region, even when scene-level RRI would prefer large background surfaces.

Model

***EFM3D* – Egocentric Foundation Model 3D**: Egocentric 3D foundation-model stack used as the frozen spatial backbone for ARIA-NBV candidate scoring. The current project uses EFM3D and its EVL architecture to expose voxel occupancy, centerness, semantic, and OBB evidence for VIN-style RRI prediction.

***EVL* – Egocentric Voxel Lifting**: EFM3D architecture that lifts synchronized egocentric observations into a gravity-aligned 3D voxel feature volume. ARIA-NBV uses EVL head outputs, pre-head features, and OBB predictions as local actor-visible evidence and target support, not as the complete frozen scene representation.

***Q__H* – Finite-Horizon Q Function**: Target-conditioned finite-candidate value family Q__theta(s,e,a,h) that scores each valid candidate for an explicit requested residual horizon h up to a configured maximum H. The thesis-core architecture is one shared horizon-conditioned scorer; dense Q__1 and exact Q__2 targets establish the recursion, horizon-indexed fitted Q supplies longer-horizon targets, and Double Q remains an optional correction for a noisy learned successor maximum.

***target-conditioned scorer* – Target-Conditioned Scorer**: VIN-style candidate scorer that receives scene state, a candidate view, and an encoding of the target of interest. The scorer predicts target-specific utility so view ranking can prioritize a selected entity or region instead of only optimizing scene-level RRI.

***VIN* – View Introspection Network**: Learned candidate-view scorer that predicts RRI or an ordinal RRI-derived utility without capturing the candidate view. ARIA-NBV adapts VIN-NBV by placing a lightweight RRI prediction head on top of frozen EVL features and candidate-pose evidence from ASE snippets.

Planning

***cost* – Acquisition Cost**: Budget consumed to acquire observations, measured by view count, path length, elapsed time, invalid-action rate, or a weighted combination. The thesis should first report cumulative root-normalized target gain, diagnostic target RRI, and acquisition cost separately, then use scalarized objectives only when the tradeoff is explicit.

candidate **– Candidate View**: Proposed camera pose whose expected reconstruction utility is evaluated before selecting the next observation. ARIA-NBV samples candidate views around a reference pose or target, renders candidate depths from the mesh for oracle labels, and scores candidates with RRI or learned VIN predictions.

transition **– Counterfactual Transition**: Deterministic or replayable rollout update that renders or retrieves only the selected candidate observation, backprojects it to candidate geometry, and fuses it into the accumulated reconstruction proxy.

action set **– Finite Candidate Action Set**: Valid discrete action set for ARIA-NBV planning, obtained by masking a sampled candidate view table. The action is the candidate-table index, not a continuous pose; the selected pose is $q_t = q_{t,a_t}$. Invalid candidates are constraints, not low-quality actions.

return **– Finite-Horizon Return**: Symbolic H-step return used by ARIA-NBV rollout evaluation and Q_H targets. The formula keeps γ explicit while the thesis-core comparison reports cumulative root-normalized target gain under equal acquisition budget.

5DoF **– Five Degrees of Freedom**: Reduced camera-action parameterization commonly used when roll is fixed or otherwise constrained. ARIA-NBV uses 5-DoF language for candidate and planning abstractions where position and viewing direction matter while roll is not an independently optimized control dimension.

CF+ state **– Geometry-Rich Counterfactual State**: Ablation actor state that extends the minimal counterfactual state with selected prior synthetic observations only: rendered depth, valid mask, backprojected point cloud, and derived local normals or support summaries.

historic state **– Historic Snippet State**: Raw actor-visible state available along the logged Project Aria/ASE trajectory before counterfactual rollout synthesis. It includes captured camera streams, calibration, timestamps, trajectory/gravity, semidense points with support or uncertainty, EVL/EFM evidence, and detected or predicted OBBs; GT mesh and GT OBBs are excluded as actor inputs.

***CF0 state* – Minimal Counterfactual Actor State**: Main thesis-core actor-visible counterfactual rollout state. It keeps the fused point proxy $\mathcal{P}_t$ as broad scene state, optional lifted image-foundation features attached to points, local root EVL evidence for target support, selected-view history, target descriptor, budget, candidate table, validity masks and reason codes, and current candidate-query features, without exposing all-candidate GT renders.

***NBV* – Next-Best-View**: Problem of selecting the next sensor viewpoint to improve an active reconstruction or inspection objective under a limited acquisition budget. In ARIA-NBV, the preferred objective is reconstruction quality improvement rather than coverage alone, with candidate views ranked by oracle or learned RRI scores.

***oracle state* – Oracle Rollout State**: Privileged state used only for ASE mesh-supervised label generation, upper-bound planning, and evaluation. It includes GT mesh and target crops, all-candidate renders, points, normals, mesh-face visibility, RRI and point-mesh error metrics, oracle scores, and GT OBBs.

***offline state* – Persisted Offline Sample State**: Compact VIN offline-store state persisted for training and diagnostics. It contains the VinSnippetView payload, candidate table and cameras, candidate count, validity or count metadata, oracle metrics as labels, optional rendered depths, compact OBBs, trajectory metadata, and selected EVL numeric fields.

***state* – Rollout State**: Family of rollout state variants used by ARIA-NBV. The actor-visible variants contain logged or replayed observations, reconstruction proxies, candidates, validity masks, target descriptors, history, and budget; the oracle variant adds privileged GT geometry and all-candidate labels for supervision and evaluation only.

***NBV MDP* – Target-Conditioned NBV MDP**: Finite-horizon Markov decision process used to define ARIA-NBV's target-conditioned rollout and fitted Q_H training contract. The actor observes only current reconstruction state, sampled candidate views, validity masks, target descriptors, history, and budget state; oracle meshes and GT crops are supervision and evaluation signals.

***mask* – Validity Mask**: Hard candidate-level feasibility mask used during candidate ranking, rollout generation, and Q_H training. The scalar mask $m_{t,i}$ gates selectable rows, while the reason code $\rho_{t,i}$ records why a row was rejected. Invalidity is represented separately from reconstruction quality and should not be encoded as the lowest RRI class.

Protocol

***GT-EVAL* – Ground-Truth Target Evaluation**: Main thesis protocol component restricting ground-truth OBBs and target mesh crops to oracle labels, target-RRI computation, matching checks, and evaluation. GT target crops are not actor-visible inputs in the main result.

***OBS-SEL* – Observed Target Selection**: Main thesis protocol component requiring target selection to use only actor-visible evidence such as predicted or tracked OBBs, class probabilities, confidence, projected area, and semidense or EVL support. Ground-truth target annotations are not visible to the selector in this protocol.

***PRED-Q* – Predicted-Target Q**: Main thesis protocol component requiring the scorer or fitted Q_H model to condition on predicted or observed target descriptors rather than ground-truth target inputs. It covers target-conditioned one-step scoring and finite-candidate Q_H selection.

Reconstruction

***MVS* – Multi-view Stereo**: Reconstruction family that estimates dense or semi-dense scene geometry from multiple calibrated views. MVS is part of the broader reconstruction context for NBV, although ARIA-NBV's current oracle labels are built from ASE meshes and Aria-style semi-dense point observations.

***PC* – Point Cloud**: Set of 3D points representing observed scene geometry. ARIA-NBV compares current and candidate-fused point clouds against ground-truth meshes to compute oracle RRI and surface-distance diagnostics.

***SLAM* – Simultaneous Localization and Mapping**: Method for estimating sensor motion while building a map of the surrounding scene. In ARIA-NBV, logged SLAM poses and semi-dense points form the current reconstruction state against which candidate-view RRI is evaluated.

***track* – Track**: Temporal sequence of corresponding image-feature detections across frames, usually carrying per-frame image coordinates, timestamps, and camera IDs. Tracks can be triangulated or optimized into 3D points, making them a bridge between image evidence and the semi-dense point clouds used by ARIA-NBV.

***VIO* – Visual-Inertial Odometry**: Pose-estimation method that combines visual measurements from cameras with inertial measurements from IMUs. Project Aria pose and mapping products build on visual-inertial estimation, and ARIA-NBV depends on those pose streams for snippet state and candidate frame contracts.

Representation

***Occupancy Grid* – Occupancy Grid**: Spatial grid whose cells encode whether space is occupied, free, unknown, or represented by a related occupancy probability. Occupancy-style voxel evidence appears in EVL outputs and VIN feature construction, where it helps summarize local 3D scene state for candidate scoring.

***3DGS* – 3D Gaussian Splatting**: Explicit radiance-field representation using optimized 3D Gaussian primitives for real-time novel-view synthesis. ARIA-NBV treats active 3DGS work as proposal, uncertainty, and simulator-bridge literature, not as a replacement for ASE mesh-supervised RRI.

Supervision

***oracle RRI* – Oracle RRI**: RRI label computed with privileged ground-truth geometry, used for supervised training and evaluation. The current oracle renders candidate depth maps from ASE ground-truth meshes, backprojects candidate point clouds, fuses them with the current semi-dense reconstruction, and scores the resulting surface-distance improvement.

List of Symbols

Symbol	Meaning	Key
	Oracle and Reconstruction	
RRI	Relative Reconstruction Improvement score for a candidate view.	`oracle.rri`
$\mathcal{P}$	Actor-visible point set accumulated from observed or replayed views.	`oracle.points`
$\mathcal{P}_q$	Candidate-view point contribution used in oracle or counterfactual updates.	`oracle.points_q`
$\mathcal{Q}$	Finite candidate-view set considered by the planner.	`oracle.candidates`
$\mathcal{Q}_t$	Candidate-view set available at rollout step t.	`oracle.candidates_t`
$q_{t,i}$	Candidate pose i at rollout step t.	`oracle.candidate_qti`
$\boldsymbol{D}_q$	Oracle-rendered or candidate-specific depth map for a proposed view.	`oracle.depth_q`
$D_{P \to M}$	Point-to-mesh directional error component.	`oracle.acc`
$D_{M \to P}$	Mesh-to-point directional error component.	`oracle.comp`
D	Aggregate point-mesh reconstruction error used by RRI definitions.	`oracle.err`
	ASE Assets	
$\mathcal{M}^{\mathrm{GT}}$	Ground-truth scene mesh used for oracle labels and evaluation.	`ase.mesh`
$\mathcal{M}_e^{\mathrm{GT}}$	Ground-truth mesh crop for target e.	`ase.mesh_target`
$\mathcal{F}^{\mathrm{GT}}$	Ground-truth mesh faces used in point-mesh distance calculations.	`ase.faces`
$\boldsymbol{T}_{\mathrm{rig}}^{\mathrm{w}}(t)$	Project Aria rig pose in the world frame at time t.	`ase.traj`
$\boldsymbol{T}_{\mathrm{rig}}^{\mathrm{w}}(T)$	Final rig pose used as an anchor for gravity-aligned local fields.	`ase.traj_final`
	VIN Scorer	
r	Target or scene RRI value used as a model target.	`vin.rri`
$\hat{r}$	Predicted RRI value emitted by the learned scorer.	`vin.rri_hat`
$\mathcal{L}$	Training loss for scorer or value-model objectives.	`vin.loss`
m	Candidate validity indicator used by scorer diagnostics.	`vin.cand_valid`
	Entity and Target	
RRI_e	Target-specific Relative Reconstruction Improvement for entity e.	`entity.rri_e`

Symbol	Meaning	Key
$\mathrm{RRI}^{e}_{t,i}$	State-relative marginal target RRI for candidate i at rollout step t.	`entity.target_rri_marginal`
$C_t^{\mathrm{RRI},e}$	Running sum of selected state-relative target RRIs.	`entity.target_rri_cumulative`
J_t^e	Running cumulative target gain normalized by rollout-root error.	`entity.target_root_gain_cumulative`
$\boldsymbol{\varphi}_e$	Actor-visible target/entity descriptor used to condition target-specific value prediction.	`entity.target_desc`
	Planning and Rollout	
s	Abstract planning state.	`rl.s`
a	Action index selecting a candidate row.	`rl.a`
r	Scalar reward or immediate gain.	`rl.r`
G	Finite-horizon return accumulated across rollout steps.	`rl.G`
$\mathcal{M}_{\mathrm{NBV}}$	Target-conditioned finite-candidate NBV decision process.	`rl.mdp_nbv`
$\mathcal{A}(s_t)$	Valid action set available in state s_t.	`rl.action_set`
T	State-transition operator for selected candidate actions.	`rl.transition`
s_t^{hist}	Logged historic snippet state at rollout step t.	`rl.s_hist`
s_t^{off}	Persisted offline sample state used by training readers.	`rl.s_off`
s_t^{cf0}	Minimal actor-visible counterfactual state.	`rl.s_cf0`
$s_t^{\mathrm{S0\text{-}pose}}$	Implemented fixed-root pose-history state used by qh_cf0_v1.	`rl.s_pose`
$s_t^{\mathrm{cf+}}$	Geometry-enriched counterfactual successor state.	`rl.s_cf_geom`
$s_t^{\mathrm{CF\text{-}GT\text{-}carrier}}$	Implemented privileged selected-depth carrier used by qh_cfplus_gt_depth_v1.	`rl.s_cf_gt_carrier`
s_t^{oracle}	Oracle-only state used for labels, upper bounds, and evaluation.	`rl.s_oracle`
r_t^e	Target-specific reward at rollout step t.	`rl.reward_target`
$G_t^{(H)}$	H-step finite-horizon return from rollout step t.	`rl.return_h`
Q_H	Finite-horizon candidate-value function.	`rl.qh`
γ	Return discount factor.	`rl.gamma`
H	Planning or rollout horizon length.	`rl.H`
$m_{t,i}$	Hard validity mask for candidate i at rollout step t.	`rl.validity_mask`
$\rho_{t,i}$	Invalid-action reason code for candidate i at rollout step t.	`rl.invalid_reason`
e_t	Selected target entity at rollout step t.	`rl.target`
b_t	Remaining acquisition budget at rollout step t.	`rl.budget`
$C(\tau)$	Acquisition cost of a selected trajectory.	`rl.acquisition_cost`

Symbol	Meaning	Key
	Shape and Size	
N_q	Number of candidate views in a candidate table.	`shape.Nq`
K	Number of ordinal bins or discrete levels when used in scorer losses.	`shape.K`
	scene	
$\boldsymbol{M}_t^{\mathrm{ray}}$	Sparse actor-visible ray-aware memory separating occupied, free, and unknown support at step t.	`scene.ray_memory_t`
	Observation and Actor State	
$\boldsymbol{F}_t^{\mathrm{DINO@pt}}$	Visibility-gated logged DINO feature bank attached to semidense or fused world points.	`obs.dino_point_bank_t`
$\boldsymbol{X}_t^{\mathrm{pt}}$	Point-token set combining geometry, support, uncertainty, history, and optional compressed logged descriptors.	`obs.point_tokens_t`
	scene	
$\boldsymbol{\Phi}_t^{\mathrm{scene}}$	Composite actor-visible scene-memory descriptor queried by the finite-candidate value model.	`scene.scene_memory_t`
$\boldsymbol{E}_0^{\mathrm{EVL\text{-}local}}$	Root local EVL evidence field used for target/candidate support reads.	`scene.evl_local`
$\omega_{t,i}^{\mathrm{EVL}}$	Candidate or target-query fraction supported by the root EVL voxel extent.	`scene.evl_support_frac`
$\boldsymbol{g}_{t,i}^{\mathrm{EVL}}$	Pooled local EVL evidence token for a target, candidate frustum, or target-candidate intersection query.	`scene.evl_support_token`
$\boldsymbol{g}_e^{\mathrm{tgt}}$	Pooled target-support descriptor for the selected target hypothesis.	`scene.target_support_pool`
$\boldsymbol{g}_{t,i}^{\mathrm{fr}}$	Candidate-frustum support descriptor pooled from actor-visible scene memory.	`scene.frustum_support_pool`
$\boldsymbol{g}_{t,e,i}^{\mathrm{cap}}$	Pooled descriptor over the intersection between target support and candidate frustum support.	`scene.target_frustum_pool`
$\boldsymbol{g}_{t,i}^{\mathrm{ray}}$	Candidate-conditioned ray query over occupied, free, unknown, hit, target-support, and uncertainty summaries.	`scene.ray_query_ti`
RenderQuery	Abstract query operator that summarizes scene memory in a candidate camera/frustum without introducing fresh counterfactual RGB features.	`scene.render_query`
	spatial	
r_t	Reference pose for candidate-relative descriptor construction at decision step t.	`spatial.ref_pose`
$\boldsymbol{T}_{r_t,i}^{\mathrm{rel}}$	Relative transform from the reference pose r_t to candidate camera i.	`spatial.ref_candidate_transform`

Symbol	Meaning	Key
$\boldsymbol{h}^{\text{pose}}_{t,i}$	Relative/local candidate pose descriptor derived from a reference pose rather than raw world coordinates.	`spatial.candidate_pose_feat`
$\boldsymbol{h}^{\text{rel}}_{t,e\|i}$	Candidate-target relation descriptor in the candidate/query local frame.	`spatial.candidate_target_rel_feat`
$\boldsymbol{e}^{\text{rel}}_{a\|i}$	Query-local relative positional embedding for target, history, support, or candidate relations.	`spatial.relation_rpe`
	model	
$\boldsymbol{h}^{\text{tgt}}_{e}$	Learned selected-target token built from the actor-visible target descriptor and scene support.	`model.target_token`
$\boldsymbol{x}_{t,i}$	Per-candidate row feature assembled from pose, relation, support, validity, provenance, and history descriptors.	`model.candidate_row`
	Entity and Target	
$J_e^{(H)}$	Endpoint reconstruction gain for target e over horizon H.	`entity.endpoint_gain`
$J_{e,\log}^{(H)}$	Log-scale endpoint reconstruction gain for target e.	`entity.log_gain`
Δ_{look}	Gain available to an oracle lookahead policy beyond one-step selection.	`entity.lookahead_headroom`
η_Q	Fraction of oracle lookahead gain recovered by the Q policy.	`entity.q_recovery`
$G_t^{(H)}$	Target-conditioned finite-horizon return.	`entity.return_h`
	Observation and Actor State	
$\boldsymbol{D}$	Depth observation sequence.	`obs.depth`
$\boldsymbol{n}$	Per-point or per-face normal observation.	`obs.face_normal`
$\boldsymbol{I}^{\text{rgb}}$	RGB image observation.	`obs.img_rgb`
$\mathcal{P}^{\text{cf}}$	Counterfactual point observation.	`obs.points_cf`
$\mathcal{P}^{\text{semi}}$	Semi-dense observed point set.	`obs.points_semi`
$\mathcal{P}_t$	Point set available at rollout step t.	`obs.points_t`
	Planning and Rollout	
$\mathcal{Q}_t$	Finite candidate-view table at rollout step t.	`rl.candidate_table`
	VIN Scorer	
$\boldsymbol{F}_v$	Learned value field evaluated at v.	`vin.field_v`

List of Figures

List of Tables

1 Introduction

Active reconstruction couples sensing and inference: after a fixed acquisition budget, reconstruction quality depends not only on the surface estimator but also on which feasible viewpoints were acquired. Classical active perception therefore treats sensing actions as part of perception, and view-planning work formalizes the recurring generate–score–select loop for three-dimensional inspection [AWB88, Baj88, SRR03]. This thesis studies a deliberately bounded instance of that problem: target-conditioned view selection from a finite admissible candidate table for egocentric indoor reconstruction.

VIN-NBV provides the closest objective precedent by supervising candidate ranking with RRI, computed from the reduction in point–mesh reconstruction error after adding a query view [Fra+25]. ARIA-NBV retains this quality-driven principle but makes four tensions explicit: coverage or greedy selection need not equal target endpoint quality; oracle quality labels need not be actor-visible inputs; learned lookahead can exceed finite offline support; and representation capacity cannot recover evidence that the egocentric stream did not provide. The planning question is therefore bounded: does oracle lookahead improve target endpoint quality over one-step selection under the same finite candidate support, and can a learned policy use only the permitted observations? Project Aria, ASE, and Egocentric Foundation Model 3D (EFM3D) provide the calibrated egocentric, mesh-supervised, and local actor-visible substrate for this study [Eng+23, Met25a, Str+24].

The current data-generation tasks are defined directly from geometry-valid GT oriented bounding boxes. They provide target identity, target crops, and oracle labels, but they do not implement actor-visible target discovery or identity matching. Consequently, deployable-input claims require a separate observed- or predicted-target protocol; the current target tasks support oracle supervision and controlled evaluation only.

1.1 Objectives and boundary

The thesis objective is a finite-candidate, target-conditioned NBV comparison with equal acquisition budgets and oracle re-evaluation. The boundary is explicit: GT geometry and meshes provide labels and evaluation only; the actor-visible policy receives observed or predicted target descriptors and validity masks. Claims are restricted to the sampled candidate support, horizon, branch factor, target protocol, and held-out split.

Problem statement. Given a GT-defined target task, actor-visible reconstruction evidence, a finite candidate table with hard validity constraints, and a fixed acquisition budget, determine whether bounded oracle lookahead improves endpoint target reconstruction over one-step oracle-greedy selection and, only if such headroom exists, whether a learned policy using non-privileged inputs recovers a meaningful fraction of it under the same oracle evaluation.

The thesis separates supervision from decision-time information. Ground-truth geometry defines current target tasks, target crops, oracle labels, and evaluation; it is not an ordinary actor input. Invalid candidates are outside the admissible action set rather than examples with low utility. One-step oracle greedy is an immediate-reward comparator over the evaluated valid candidate table, while oracle lookahead is an upper reference only within the fixed candidate generator, horizon, branch factor, target pool, and validity regime. These restrictions make negative results interpretable without turning a bounded experiment into a universal statement about view planning.

1.2 Research Questions

The thesis asks whether ARIA-NBV can perform target-conditioned, quality-driven multi-step NBV (NBV) selection with a finite candidate action space. The utility is target-specific RRI (RRI), and every learned action is re-evaluated with the oracle under the same candidate support, validity rules, and acquisition budget. GT (GT) geometry defines target tasks, labels, and evaluation; it is not an ordinary actor input.

1.2.1 Objectives and scope

The work has six linked objectives. First, validate the geometry, candidate-label, invalidity, split, and oracle-RRI contracts on a trusted ASE subset before scale-up. Second, define the observed-target-selection / predicted-target-Q / GT-evaluation protocol. Third, establish a learned one-step target-conditioned scorer as the required myopic control. Fourth, validate mixed candidate sets and rollout support before training a finite-horizon value model. Fifth, train $Q_{H,\theta}$ from ASE oracle rollouts as a finite-candidate value model, first using candidate-to-state queries. Finally, grow support from trusted subsets to scene-level ASE evidence only while preserving mesh-supervised target RRI.

Online discrete Q_H is a deferred RQ5 bridge; continuous or simulator-backed actor–critic control is RQ6 and is not required for the finite-candidate thesis result. The active quantitative core is therefore RQ1–RQ4 below.

Development provenance and reconciliation. The reviewed Quarto source (the reviewed historical QMD at `origin/main`) supplied the six-tier ordering: RQ1 objective, RQ2 offline finite-candidate planning, RQ3 actor-visible target representation, RQ4 candidate/rollout/scale support, RQ5 online discrete Q_H, and RQ6 continuous or simulator escalation. The base Typst formulation on this branch had consolidated the first four questions but omitted the two explicitly gated extensions. The current formulation preserves the reviewed ordering, makes RQ1–RQ4 the evaluated core, and states RQ5/RQ6 as conditional bridges. The Quarto page remains historical review input; this section and the executable Python/configuration owners define current terminology and implementation status.

Implementation: planned **Evidence: pending**

The actor-visible descriptor and deterministic 3D-IoU matching below are a planned hypothesis and acceptance protocol, not an implemented result.

Promotion gate: Observed/predicted target matching and target-conditioned scoring remain future work; the current oracle-task protocol is the available baseline

1.2.2 RQ1 — Objective and endpoint contract

Can target-conditioned, finite-candidate NBV improve endpoint quality for a selected target under a fixed acquisition budget while keeping training return, endpoint evaluation, and acquisition cost separate? For a rollout rooted at s_0 with target e, let $\mathcal{P}_t$ be the accumulated fused points and $\mathcal{M}_e^{\mathrm{GT}}$ the matched target surface. The oracle-only target crop gives

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

The primary fixed-budget metric is endpoint gain J_H^e. The default multi-step reward is root-normalized target gain,

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

$$G_{t,e}^{(h)} = \sum_{k=0}^{\min(h,b_t)-1} \gamma^k r_{t+k}^e$$

State-relative target RRI remains a diagnostic and VIN-compatible one-step label. We report both its marginal and selected-chain cumulative forms,

$$\mathrm{RRI}_{t,i}^e = \frac{\Delta_t^e - \Delta_{t|i}^e}{\max(\Delta_t^e, \varepsilon)}$$

$$C_t^{\mathrm{RRI},e} = \sum_{k=0}^{t-1} \mathrm{RRI}_{k,a_k}^e$$

without treating their non-telescoping sum as endpoint gain. Log-error gain and motion or feasibility costs are explicit ablations. Invalid candidates are hard constraints, never the lowest-RRI class. Report endpoint gain, cumulative root gain, diagnostic target RRI, scene RRI, view count, path length, invalid-action rate, and runtime at matched budgets.

1.2.3 RQ2 — Offline finite-candidate planning

Does bounded oracle lookahead have positive endpoint-quality headroom over one-step oracle-greedy selection, and can an offline $Q_{H,\theta}$ policy recover a measurable fraction of it? The first comparison fixes target tasks, roots, candidate support, validity, horizon, and budget. Define

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

If this headroom is indistinguishable from zero for the evaluated support, the result is a setup-specific negative finding, not a claim that target RRI is universally myopic. When it is positive, $Q_{H,\theta}$ is trained from ASE oracle rollout traces and compared with random-valid, one-step oracle greedy, learned one-step scoring, bounded oracle lookahead, and oracle-scored temperature-softmax traces. The learned model emits one masked bounded-horizon value per finite candidate; GT meshes, crops, and oracle values remain supervision/evaluation only. Gumbel-Top-k is later diversity evidence, not a prerequisite for the first run.

1.2.4 RQ3 — Actor-visible target representation

Which actor-visible target descriptor and matching protocol support one-step and finite-horizon scoring without privileged target geometry or all-candidate oracle renders at decision time? The proposed protocol separates observed or predicted target selection, predicted-target conditioning, and ground-truth target-crop evaluation. The first descriptor contains observed/predicted Oriented Bounding Box (OBB) centre, extents, orientation, class, confidence, projected area, relative pose, semi-dense point support, and EVL support. A compact crop descriptor from actor-visible spatial evidence is the first ablation; entity tokens and appearance features are later ablations. A ground-truth target box is retained only as a privileged sanity or upper-bound path, not as an actor-visible deployment input.

The planned matcher first requires compatible semantic class, then applies deterministic 3D IoU with an explicit threshold τ_{IoU}. If the best and second-best compatible matches are separated by less than the explicit ambiguity margin τ_{gap}, the proposal is rejected as ambiguous; below-threshold matches are unmatched. Visibility, projected area, and semi-dense/EVL support are audit-only evidence fields for this association rule, not association scores. Unmatched or ambiguous targets are protocol-invalid cases, not low-RRI examples.

1.2.5 RQ4 — Candidate, rollout, and scale support

Do mixed target-centric and default-exploration candidates, controlled rollout branching, and scale-aware replay evidence improve reliability over a narrower candidate family? The thesis-core finite table is the executable 60-row mixture: forward-local (24), target-bearing-local (24), and lateral-target-bypass (12). Legacy radial-away/radial-towards and unrestricted free-shell directions are explicit ablations, not thesis-core families. Every row retains strategy provenance, a hard validity mask, and explicit invalid-reason codes. Branch, beam, and stochastic temperature controls widen data support; they do not replace the RRI objective.

Scale proceeds causally: a small trusted subset first establishes geometry, matching, invalidity, and replay correctness; a scene-level held-out or full ASE GT-mesh subset then supplies thesis evidence; external substrates are eligible only when they preserve target-specific mesh/oracle supervision. Report scenes, snippets, targets, trajectories, anchor poses, candidates, rollout seeds, transitions, split boundaries, invalid gaps, and missing coverage separately. Sample-level splits across snippets from one scene are not sufficient for final claims.

1.2.6 Shared evidence protocol

Invalidity is a shared constraint across all questions. Only physical or explicit admissibility failures (for example collision, out-of-bounds pose, bad frustum, or a contract-defined motion violation) clear action validity; outside-EVL extent, missing semidense/appearance support, and similar evidence gaps remain support fields unless the executable feasibility contract says otherwise. A failed oracle evaluation does not make the action physically invalid: it clears oracle-label validity and Q-training eligibility and is reported as a label/evaluation failure. Keep the mask taxonomy explicit: **action validity** gates selectable candidate rows; **actor evidence/support** records whether observed target or local semidense/EVL evidence exists; **target validity** records whether the target task and identity state are admissible; **oracle-label validity** records whether ground-truth evaluation produced a finite label; and **Q-training eligibility** is the strict intersection required by the learning contract. Report invalid fractions and reason distributions, target-visible fractions, masked and unmasked rank metrics, and selected-action oracle evaluation. Equal budget means equal horizon, candidate count, candidate distribution, and validity constraints; path length, runtime, and oracle calls are separate measurements. Coverage, calibration, stage shift, and storage lineage are evidence qualifiers rather than proxy objectives.

1.2.7 RQ5 — Conditional online discrete Q_H bridge

After offline finite-candidate evidence is stable, does online interaction over the same discrete candidate contract improve endpoint target gain or calibration over the offline Q_H policy? RQ5 reuses the actor-visible state, candidate mixture, hard action mask, target protocol, horizon, and oracle re-evaluation of RQ1–RQ4. It is attempted only when offline headroom and replay support are positive; it is not required to establish the finite-candidate thesis result. Online training must not silently expand the action space, change the target protocol, or treat GT renders as actor evidence.

1.2.8 RQ6 — Lower-priority continuous and simulator escalation

If the finite-candidate and online-discrete evidence is stable, does a continuous or hierarchical target-then-pose policy, potentially in a simulator, provide measurable headroom over the best finite-candidate policy under the same target-specific objective? RQ6 is a time-permitting escalation, not a substitute for the offline Q_H result. Any comparison must preserve target conditioning, matched budgets or cost curves, explicit feasibility handling, and independent oracle endpoint evaluation; no continuous-control result is implied by the current finite-candidate implementation.

Validation TODO: Freeze one claim-level contribution statement for each research question and link it to the exact estimand, evidence artifact, and decision rule that can answer it. Until those links exist, the questions define intended scope rather than validated contributions.

Source: Chapter 5; Chapter 6

Gate: a complete RQ-to-estimand-to-result ledger backed by confirmatory artifacts

1.2.9 Research matrix

Question	Gate	Primary evidence

RQ1	M1, M5	Endpoint target gain, cumulative root gain, diagnostic RRI, and acquisition-cost curves.
RQ2	M4, M5	One-step gate, oracle lookahead, rollout traces, endpoint gain, and recovered headroom.
RQ3	M3, M4	Actor-visible target encoding and leakage-safe crop ablation.
RQ4	M2–M7	Candidate mixtures, branching diversity, validity statistics, scale, and coverage reports.
RQ5	M6, conditional	Online discrete Q_H under the unchanged finite-candidate contract.
RQ6	M6, lower priority	Continuous or simulator escalation against the best discrete policy.

The matrix is a reporting map, not a second source of contracts: implementation semantics remain owned by code, tests, and the linked development reports.

2 Foundations

ARIA-NBV combines ASE trajectories and GT geometry, local EFM3D evidence, oracle RRI labels, and VIN-style finite-candidate scoring. ASE supplies calibrated Aria-like streams and aligned semi-dense maps; EVL supplies frozen local voxel evidence rather than a complete long-horizon scene memory [Met25a, Met25b, Str+24]. GT OBBs currently define data-generation target tasks, whereas observed or predicted target discovery remains outside the implemented actor path.

2.1 Related Work

Next-best-view work exposes a tension between cheap coverage-oriented selection and endpoint reconstruction quality. PB-NBV makes the finite generate–score–select abstraction explicit: it samples candidate viewpoints because direct 6-DOF optimization is difficult, then evaluates candidates with a projection-based quality function and reports point-cloud coverage and computation time [Jia+25]. VIN-NBV keeps the same sequential greedy candidate-selection structure for its coverage baseline, but replaces the coverage score with predicted Relative Reconstruction Improvement and evaluates reconstruction quality with point-level error [Fra+25]. These results motivate a matched comparison, not a universal preference: ARIA-NBV treats coverage or one-step quality as a comparator and asks whether target-specific endpoint quality changes under the same finite candidate support.

Quality supervision creates a second tension: the label can use information that the actor must not receive. VIN-NBV defines an oracle variant by replacing the learned score with ground-truth RRI, making the distinction between a quality label and a decision-time observation explicit [Fra+25]. EFM3D supplies the complementary egocentric substrate: Aria sequences, oriented-box metadata, calibrated spatial annotations, and ground-truth meshes support controlled supervision and evaluation [Eng+23, Str+24]. The resulting separation is useful for ARIA-NBV, but it does not solve target discovery: a geometry-defined target task can provide an oracle crop and label while a deployable actor would still need an observed or predicted target record.

Offline value learning exposes a third tension between a finite behavior-supported dataset and a learned lookahead maximum. CQL identifies distributional shift and bootstrapping from out-of-distribution actions as sources of optimistic value estimates, and its conservative objective is tied to the dataset action distribution [Kum+20]. BCQ makes the same boundary operational by constraining selected actions toward those present in the batch [FMP19], while Double DQN separates action selection from action evaluation to reduce maximization bias [HGS15]. For ARIA-NBV, these works support a bounded methodological question rather than a result: a learned finite-horizon scorer must be checked against logged candidate support and endpoint re-evaluation, because an extrapolated high value is not evidence of a better target endpoint.

Representation capacity creates a fourth tension with the bounded evidence available to an egocentric actor. Deep Sets and Set Transformer provide permutation-aware processing for unordered candidate sets [Lee+19, Zah+17], while Point Transformer V3 and Minkowski sparse convolutions show two ways to increase point-cloud receptive field or compute only on occupied coordinates [CGS19, Wu+24]. EGNN adds translation, rotation, and permutation equivariance to coordinate-bearing graph features [SHW21]. These mechanisms address order, geometry, or efficiency, but they do not supply observations that were never logged: in ARIA-NBV, richer encoders remain conditional ablations whose value must be established under the same actor-visible evidence, finite candidate table, and endpoint evaluation.

2.2 Geometric Learning and Candidate-Set Theory

Remove or rewrite before submission: Separate method-independent requirements that the final scorer must satisfy from speculative representation ladders and curricula. Retain only theory that is tested by the frozen method or needed to interpret its failures.

Source: this section and Chapter 4

Gate: one implemented scorer contract and matching acceptance tests determine the surviving theory

The learned object in ARIA-NBV is a typed finite-candidate decision map: given a target record, selected history, partial actor-visible geometry, and an unordered table of feasible candidate poses, assign one score to each candidate for selection under later oracle re-evaluation. The model must respect the symmetries and information boundary of this map before architectural capacity is interpreted [Bro+21].

The first required structure is candidate-row permutation equivariance. Reordering $\mathcal{Q}_t$ may reorder per-candidate Q_H outputs, but it must not change the value assigned to the same physical candidate. Deep Sets supplies a pooled symmetry baseline, while Set Transformer supplies candidate interaction without assigning semantic meaning to row order [Lee+19, Zah+17]. Both retain a row-level scoring path because the policy selects a candidate row.

For any permutation matrix Π acting on candidate rows, the value model should satisfy

$$f_\theta(\Pi \boldsymbol{X}_t, \Pi \boldsymbol{m}_t) = \Pi f_\theta(\boldsymbol{X}_t, \boldsymbol{m}_t)$$

where $\boldsymbol{X}_t = \left\{\boldsymbol{x}_{t,i}\right\}_{i=1}^{N_q}$ stores per-candidate actor-visible descriptors and $\boldsymbol{m}_t$ stores validity. Invalid and padded rows are outside the admissible action set, not low-utility examples. Selection is therefore

$$\mathcal{A}_t = \left\{i \in \left\{1, ..., N_q\right\} : m_{t,i} = 1\right\}, \quad a_t^\theta = \underset{i \in \mathcal{A}_t}{\operatorname{argmax}} f_{\theta,i}(\boldsymbol{X}_t, \boldsymbol{m}_t)$$

The second required structure is frame discipline. Candidate, target, current-camera, and history poses use reference- or query-local relative features such as $\boldsymbol{T}_{c_q}^{r}$ and continuous rotation encodings [Zho+19, Zho+23]. This prevents an arbitrary world origin or row order from becoming a shortcut while preserving gravity alignment and camera-frustum geometry. It is a controlled coordinate choice, not a claim of full global SE(3) invariance.

The third required structure is explicit selected-view history. A camera pose also defines a viewing direction from which target-local surfaces may or may not have been observed. A compact history on $\mathbb{S}^2$ can therefore distinguish directional novelty from generic pose distance [e3n25]. Coverage, overlap, and uncertainty remain diagnostic features unless paired oracle evaluation shows that they improve target-specific RRI [GML22, JLD24].

Object	Required structure	Testable consequence
Candidate rows	Permutation-equivariant per-row value map.	Row-shuffle tests must satisfy $f_{\theta(\Pi X, \Pi m)} = \Pi f_{\theta(X,m)}$ for every selection score.
Invalid and padded rows	Separate feasibility head, trained on every non-padding row; RRI/value loss only on rows certified valid. Hard action masking is the initial policy, with a later calibrated soft feasibility gate.	Invalid rows cannot change valid-row scores except through explicit valid-count or support features.
Candidate-target geometry	Target-local and candidate-local relative pose features.	World-frame origin, yaw convention, and display-only camera transforms cannot become shortcuts.
Selected-view history	Directional memory on $\mathbb{S}^2$.	Novelty tests separate already-observed target directions from generic pose distance.
Actor/oracle boundary	Typed provenance for target descriptors, labels, and support features.	GT-defined tasks, meshes, crops, and all-candidate renders supervise data generation and evaluation only.

Table 1: Minimum geometric-learning contract for the finite-candidate value model.

The implemented hard mask defines $m_{t,i} = 0$ as exclusion from the action set, not as a numerical return. An invalid row's RRI is undefined: it must be neither forced to zero nor assigned a synthetic negative target, because zero can be a legitimate valid-row RRI and either choice would conflate feasibility with quality.

Implementation: planned **Evidence: pending**

A separate feasibility head is planned. It is not part of the current scorer or training module.

Literature: [GE19, Liu+19]

Promotion gate: implement feasibility head and evaluate held-out calibration without weakening the hard mask

Development source: aria_nbv/aria_nbv/rollouts/zarr_store.py; planned finite-horizon scorer

For each non-padding row, the planned model emits a feasibility probability and an RRI/value estimate. Binary cross-entropy supplies negative evidence on invalid rows, while the RRI loss is evaluated only when $m_{t,i} = 1$. This factorization treats feasibility as a reject decision rather than an artificial low-RRI target.

The hard mask remains the deployment safety constraint. A soft feasibility score is only a training and ranking ablation: after held-out feasibility calibration, compare a score proportional to predicted validity times the conditional valid-row value against the hard-mask baseline, while retaining the hard mask wherever it is available. The abstention literature likewise models rejection as a separate outcome rather than a continuous-regression target [Liu+19]. The curriculum hypothesis is therefore to train feasibility from the first batch, warm up the value head only on oracle-valid rows, and then introduce the calibrated soft-gate ablation [Ben+09]. It must never reintroduce an RRI target or RRI gradient for invalid rows.

Research TODO: Compare hard masking against a calibrated feasibility-times-value ranking only after feasibility calibration is measured. The hard mask remains the safety constraint in every comparison.
Source: invalid-row supervision hypothesis
Gate: feasibility calibration and policy ablation

These requirements become replay-field and model acceptance tests in Chapter Section 4.4. Row shuffles must permute outputs, invalid-row perturbations must leave valid scores unchanged, and equivalent local-frame encodings must preserve physical candidate identity. Architectural comparisons are interpretable only after those contracts hold.

The resulting foundation is narrow: active perception motivates action-conditioned sensing, VIN-NBV supplies the quality-driven one-step precedent, ASE and EFM3D define the logged egocentric evidence boundary, and finite-action learning supplies mask and support controls. Coverage and uncertainty remain diagnostics rather than substitutes for target-specific reconstruction quality.

3 Oracle and Data Generation

This chapter defines the non-deployable pipeline that turns logged ASE snippets into supervised target-conditioned NBV tasks. It separates actor state from privileged data-generation state, specifies target-task and candidate construction, defines hard invalidity and target-specific RRI, and records the resulting selected counterfactual chains. The learned method in Chapter 4 may consume only the actor-side projection of these artifacts; GT geometry, counterfactual renders, labels, and oracle search remain instruction, supervision, or evaluation assets.

3.1 State and Visibility Boundary

> **Remove or rewrite before submission:** Replace internal protocol and DTO vocabulary such as `CF-GT`, `V1`, and storage-shape terminology with scientific concepts in the main argument. Keep exact identifiers only where they provide an auditable implementation anchor.
>
> Source: this section; aria_nbv/aria_nbv/targets/protocol.py; aria_nbv/aria_nbv/data_handling/qh_data/views.py
>
> Gate: a reader can understand the information boundary without repository-local names

> **Validation TODO:** Attach exact local-source locators to the ASE and EFM3D claims, including which logged fields are observations and which geometry products are privileged supervision.
>
> Source: docs/literature/tex-src; docs/references.bib
>
> Gate: claim-level provenance comments resolve every literature claim in this section

The learned component in this thesis is a masked finite-horizon candidate-value function: it predicts Q_H values for a finite candidate table, and the decision rule selects among hard-valid rows. Training evidence is generated by offline, mesh-supervised counterfactual replay. ASE contributes both sensor-like logged streams and privileged ground truth; EFM3D converts the logged subset into a local 3D representation, whereas depth, segmentation, GT boxes, meshes, counterfactual renders, and reconstruction labels remain supervision or evaluation channels [Met25a, Str+24]. Storage in the same adapted snippet does not make these channels equally observable.

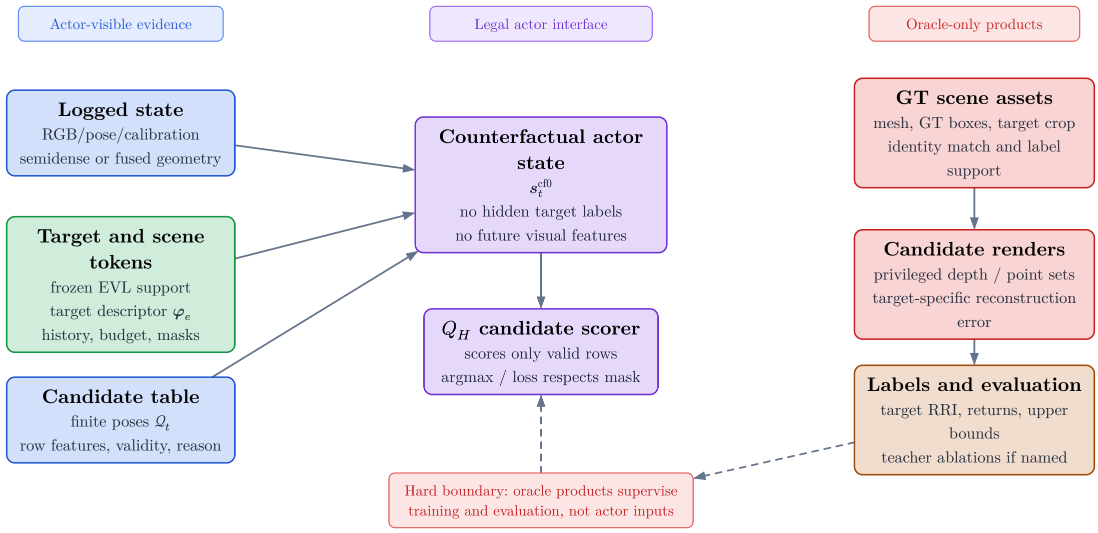


Figure 1: Actor and oracle boundary. Legal Q_H inputs are calibrated logged image evidence, poses, semi-dense geometry with uncertainty and observation support, frozen EVL features or predictions, an explicitly sourced target instruction, selected-view history, remaining budget, candidates, and masks. Reason codes remain audit evidence rather than scorer inputs. GT depth, segmentation, boxes, meshes, target crops, counterfactual renders, labels, and endpoint evaluation remain privileged.

The visibility boundary is protocol-relative and temporal. A dense render for an unselected candidate at the current decision step is oracle evidence. A render from an already selected action may enter a later state only under an explicitly named source protocol: privileged mesh-rendered depth, a declared sensor simulation, or an actor-visible observation. The same array shape can therefore denote different information, and source role must remain explicit.

3.1.1 ASE/EFM Logged Observation State

The target is treated as external task context, so the value is conceptually $Q(s_t, e, a)$ rather than a target-independent state value. The logged historic state contains the recorded trajectory evidence,

$$s_t^{\mathrm{hist}} = \left(\boldsymbol{I}^{\mathrm{rgb}}, \boldsymbol{X}, \mathcal{P}^{\mathrm{semi}}, \boldsymbol{F}_v, e_t, b_t\right) \in \mathcal{S}^{\mathrm{hist}}$$

Each component of this tuple has a distinct information role:

Calibrated multi-camera image snippets. EFM3D consumes posed and calibrated RGB and greyscale Aria streams. For every stream and time step, EVL encodes the image with a frozen 2D foundation model, projects the centers of a local 3D voxel grid into the distorted camera model, bilinearly samples the feature maps inside the camera's valid radius, and aggregates the samples across cameras and time by their mean and standard deviation [Str+24]. In the repository, `EfmCameraView` binds each normalized frame tensor to `CameraTW` calibration, device-clock timestamp, frame identifier, and an optional separately timestamped distance map. The ASE distance map is GT and remains oracle-only [Met25a]. Thus $\boldsymbol{I}^{\mathrm{rgb}}$ in s_t^{hist} denotes calibrated, time-indexed image evidence; the concrete adapted snippet also carries left and right greyscale SLAM streams rather than a single unposed RGB tensor.

Trajectory, calibration, and gravity. The adapted trajectory is a time-stamped sequence of world-from-rig `PoseTW` transforms with a world-frame gravity vector. EFM3D uses the most recent RGB rig pose, gravity-aligns its rotation, and places a local voxel grid in front of the rig; the per-camera calibration and poses then define the 3D-to-image sampling map [Str+24]. These fields therefore determine the spatial registration, metric scale, gravity direction, and temporal alignment of all lifted evidence. They are also the basis for expressing a candidate relative to the current rig rather than learning values from arbitrary world coordinates.

Semi-dense geometry and observation support. ASE provides MPS-style world-frame semi-dense points, per-point distance and inverse-distance uncertainty, timestamps, and point-to-image observation records [Met25a]. The local `EfmPointsView` preserves padded per-time point slices, valid lengths, metric uncertainty, timestamps, and spatial bounds. EFM3D voxelizes the points into a binary surface mask and uses their observing camera centers to sample the rays before each surface point into a separate free-space mask [Str+24]. Consequently, $\mathcal{P}^{\mathrm{semi}}$ alone denotes sparse surface support; free-space evidence is justified only when observation lineage and camera poses are retained. A missing point is not evidence that a voxel is free.

Derived EVL field and heads. EVL concatenates the lifted image features with the semi-dense surface and free-space masks, processes the volume with a 3D U-Net, and applies task heads for occupancy and 3D OBB detection. The paper's detection head predicts voxel centerness, class scores, and gravity-aligned box geometry; its surface head predicts occupancy and is supervised from GT depth samples labelled as free, surface, or occupied [Str+24]. The resulting $\boldsymbol{F}_v$ is therefore a learned, locally supported statistic of the logged snippet, not another raw sensor modality. The reported EFM3D configuration uses a finite 4 m × 4 m × 4 m grid anchored at the latest rig pose, so an out-of-extent target or candidate has missing EVL support rather than a measured zero or an invalid pose [Str+24].

The remaining tuple elements are decision context rather than ASE/EFM sensing modalities. e_t identifies the entity whose reconstruction is valued, and b_t conditions the return on the remaining acquisition budget. The current oracle target values are derived from a selected GT box; a deployable target must instead come from EVL predictions or another actor-visible association path. This provenance distinction is consistent with `EfmSnippetView`, which exposes camera/calibration, trajectory, and semi-dense fields as actor observations while keeping `ARIA_OBB_PADDED`, nested `gt_data`, and attached ASE meshes on the oracle/evaluation side.

3.1.2 Counterfactual Actor State

The conceptual counterfactual actor state retains the immutable logged substrate and adds only information causally available after selected actions,

$$s_t^{\mathrm{cf0}} = \left(\boldsymbol{F}_v^{\mathrm{root}}, \mathcal{P}_t, \mathcal{Q}_t, m_{t,i}, \rho_{t,i}, e_t, b_t\right) \in \mathcal{S}^{\mathrm{cf0}}$$

For the canonical model, this compact tuple is read together with the explicit ordered selected-view history:

$$\boldsymbol{h}_{t,e,i} = \mathrm{Read}\left(\boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{h}_e^{\mathrm{tgt}}, q_{t,i}, \boldsymbol{H}_t, t, H\right)$$

In s_t^{cf0}, the root EVL field remains fixed, whereas the accumulated point set $\mathcal{P}_t$ may grow with selected observations. The finite candidate table $\mathcal{Q}_t$ contains poses and generator provenance, not observations from those poses. $m_{t,i}$ defines the admissible action set, and the ordered history $\boldsymbol{H}_t$ retains the selected approach sequence. Invalid-reason codes remain audit evidence and are not scorer inputs.

The implemented profile name `qh_cf0_v1` maps exactly to the weaker `S0-pose` carrier, not to this geometry-updated conceptual state:

$$s_t^{\text{S0-pose}} = \left(\mathcal{S}_{\text{root}}^{\text{VIN}}, \boldsymbol{F}_v^{\text{root}}, \left(\boldsymbol{T}_{r,e}, \boldsymbol{l}_e\right), \mathcal{Q}_t, m_{t,i}, \boldsymbol{H}_t^{\text{pose}}, b_t\right)$$

Its actor DTO carries the immutable VIN snippet and compact root EVL evidence, target pose and extent, current candidate poses and action mask, factual selected-pose prefix, and remaining budget. It carries neither the accumulated oracle point set nor invalid-reason codes.

The richer geometry-updated state remains the planned fusion target:

$$s_t^{\text{cf+}} = \left(s_t^{\text{cf0}}, \left(\boldsymbol{D}, \boldsymbol{V}, \mathcal{P}^{\text{cf}}, \boldsymbol{n}\right)_{1:t}\right)$$

The implemented privileged `qh_cfplus_gt_depth_v1` profile stops at a causal selected-depth carrier:

$$s_t^{\text{CF-GT-carrier}} = \left(s_t^{\text{S0-pose}}, \left(\boldsymbol{D}, \boldsymbol{V}, \mathcal{K}, \boldsymbol{T}_{\text{root<-cam}}\right)_{1:t}^{\text{sel}}\right)$$

Selected $\boldsymbol{D}$ is rendered from $\mathcal{M}^{\text{GT}}$ and paired with its validity mask, candidate-camera calibration, and root-from-camera pose. The implementation does not yet fuse that carrier into $\mathcal{P}^{\text{cf}}$, free-space rays, $\boldsymbol{n}$, or a refreshed EVL field. Those are planned geometry-state derivations. Even after such fusion, the carrier contains none of the RGB or greyscale images required by EFM3D's 2D feature encoder and therefore cannot support a fresh EVL field at the selected pose [Str+24].

The transition from t to $t+1$ may use only the observation produced by the selected row a_t. It may fuse its surface and ray evidence and update support, uncertainty, direction, and recency. It may not attach image features, detections, or EVL descriptors unless the protocol also provides the calibrated camera streams from which EFM3D derives them. Counterfactual geometry must therefore retain a source role and explicit absence masks for unavailable image-derived channels.

The counterfactual state is thus an **information state** for decision making, not automatically a complete Markov state of the physical scene. It becomes task-sufficient only if its retained root context, causal dynamic evidence, explicit history, target context, action support, and budget preserve every distinction needed to predict future target-specific return. `qh_cf0_v1` is the deliberately weaker `S0-pose` baseline; `qh_cfplus_gt_depth_v1` is a privileged depth carrier, not yet a realization of the geometry-updated state.

3.1.3 Privileged Oracle State

The oracle state adds complete or counterfactual quantities outside the actor input graph:

$$s_t^{\text{oracle}} = \left(s_t^{\text{cf+}}, \mathcal{M}^{\text{GT}}, \mathcal{M}_e^{\text{GT}}, \boldsymbol{D}_q, \mathcal{P}_q, \text{RRI}\right) \in \mathcal{S}^{\text{oracle}}$$

$$\mathcal{M}_{\text{NBV}} = \left(\mathcal{S}^{\text{hist}}, \mathcal{S}^{\text{cf0}}, \mathcal{S}^{\text{oracle}}, \{\mathcal{A}_t\}, T, r_t^e, \gamma, H\right)$$

ASE provides per-frame metric GT depth aligned with RGB, per-pixel instance identifiers, class mappings, and the scene trajectory; the EFM3D release adds ASE OBB metadata and validation meshes for object-detection and surface-reconstruction supervision [Met25a, Str+24]. EFM3D uses the depth channel to supervise occupancy at sampled free, surface, and behind-surface points, while visible OBBs supervise centerness, class, and box geometry [Str+24]. These uses establish label provenance, not actor observability.

For ARIA-NBV, $\mathcal{M}^{\text{GT}}$ is the privileged reference surface used for candidate rendering and reconstruction evaluation, while $\mathcal{M}_e^{\text{GT}}$ fixes the selected entity's evaluation support. All-candidate depth $\boldsymbol{D}_q$ and backprojected point sets $\mathcal{P}_q$ answer counterfactual queries for unselected rows. RRI then compares target reconstruction error before and after adding that candidate evidence. Depth, points, crops, and RRI are legal for label generation, oracle search, diagnostics, and named upper bounds; none is a contemporaneous input to the student value model.

A GT OBB may define the current oracle identity, pose, extent, crop, and target-bearing instruction. It does not show that the actor detected or associated that object. A deployable protocol must obtain the target hypothesis from EVL predictions or another observation-derived path, retaining ASE identity, segmentation, OBBs, and meshes only for matching and evaluation.

The state hierarchy therefore separates **what was logged**, **what a selected counterfactual history may causally add**, and **what only the oracle can know**. Its layers are related by controlled information projections, not by freely copying arrays between stores. In particular, an oracle tensor may be persisted beside actor tensors for efficient replay without becoming part of the actor state.

3.1.4 Visibility, Feasibility, and Utility

Visibility answers whether a modality contains evidence for a spatial element from a particular view; feasibility answers whether an action is admissible; utility answers how beneficial an admissible action is for the target. These concepts must not be collapsed. A target can be weakly visible from a geometrically valid candidate, a candidate can lie outside EVL support while remaining physically reachable, and an oracle render can fail even though the candidate pose itself is feasible.

Invalidity is a constraint rather than a reward value. Geometry-invalid candidates receive a hard action mask and a persisted candidate reason code before policy selection, stochastic normalization, loss construction, and bootstrap maximization. A geometrically feasible candidate may still have low or negative target gain. Failure of the separate oracle evaluation does not create a candidate reason code: depending on the configured recipe, the affected row or table is skipped, or its oracle-label validity and Q-training eligibility are cleared and the oracle failure is reported separately. Oracle-derived feasibility must be distinguished from a deployable validity estimate when a policy moves from privileged target tasks to an actor-visible target protocol.

3.2 Data Generation and Target-Specific RRI Labels

Validation TODO: Validate the target-cropped error, root normalization, repeatability, failure handling, and endpoint estimand before describing RRI as a reliable study objective. Cite the exact VIN-NBV source lines for inherited definitions and distinguish every ARIA-NBV modification.

Source: docs/literature/tex-src/arXiv-VIN-NBV; metric-validation artifacts

Gate: frozen metric protocol, repeatability table, and claim-level source locators

An oracle target task fixes the entity identity and GT evaluation crop. A deployable target descriptor would instead be predicted from actor-visible observations. The current rollout generator has only the first contract: it projects a selected GT box into a compact instruction for candidate generation. Consequently, the present target descriptor denotes the requested object and its geometry; it does not establish actor-visible target discovery.

The general descriptor notation remains

$$\boldsymbol{\varphi}_e = \mathrm{Enc}_{\mathrm{tgt}}\Big(\hat{\boldsymbol{B}}_e, \hat{\boldsymbol{y}}_e, \hat{\pi}_e, A_e^{\mathrm{proj}}, n_e^{\mathrm{semi}}, n_e^{\mathrm{EVL}}, \omega_e^{\mathrm{EVL}}, \ell_e^{\mathrm{src}}, \boldsymbol{T}_{r_t,e}, \boldsymbol{T}_{c_t,e}\Big)$$

The implemented oracle sampler populates only GT identity, class, confidence, pose, extent, and reference-relative geometry. Projected area, semidense support, EVL support, and proposal-match scores are not measured by this path and must not be interpreted from their schema placeholders.

The finite action interface consists of a candidate table $\mathcal{Q}_t$, hard validity mask $\boldsymbol{m}_t$, and invalid-reason vector $\boldsymbol{\rho}_t$. The admissible set is $\mathcal{A}_t = \big\{i : m_{t,i} = 1\big\}$. Invalid rows remain logged for coverage and failure analysis but lie outside policy argmax, sampling, loss targets, and bootstrap maximization. Low RRI is a valid low-utility outcome; it is never an encoding for infeasibility.

For target e, the oracle computes a target-cropped point–mesh error Δ_t^e. Candidate selection and Q_H supervision use root-normalized target gain; state-relative RRI is retained as a diagnostic. Fixed-budget endpoint gain is the intended policy estimand, but the current replay store records cumulative selected-chain gains rather than an independent post-horizon endpoint reconstruction for every policy. Confirmatory policy comparisons must therefore add matched oracle endpoint re-evaluation or an explicitly persisted endpoint record.

3.2.1 Oracle Label Provenance

The one-step scene-level oracle from the seminar work supplies the rendering and point–mesh substrate [Fra+25, Met25c]. The target-specific pipeline renders valid candidates from the GT mesh, backprojects their depth, fuses candidate points with the privileged current evaluation cloud, crops both points and mesh to the selected target box, and evaluates the resulting point–mesh error. In the current protocol the root evaluation cloud is reconstructed from ASE GT depth, so both the crop and the root metric are oracle-only.

Camera geometry is therefore useful here only as part of a relation: a logged trajectory, a target crop, a finite candidate set, or a rendered oracle query. The calibrated camera boundary is not a contribution by itself. The consolidated candidate-support figure in Figure 3 uses camera glyphs only to expose those relations; native `CameraTW` unprojection and world-from-camera `PoseTW` remain the geometry owners behind the render path.

3.2.2 Target Selection

The implemented sampler does not match actor proposals to GT objects. It enumerates the non-padding GT OBB rows in a snippet, accepts rows with finite positive geometry, and applies seeded uniform sampling without replacement up to the configured per-snippet cap. Thus the stored `matched` status currently means geometry-valid GT task row, not successful proposal-to-identity association. IoU, ambiguity-gap, visibility, and support thresholds are absent from this admission rule.

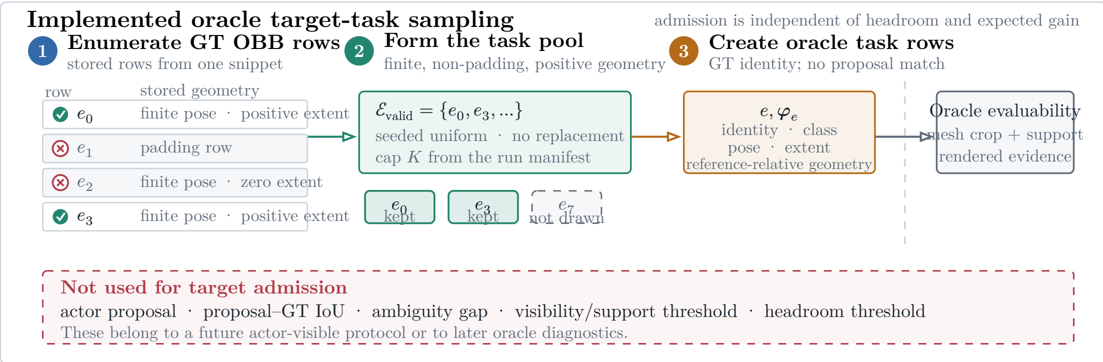


Figure 2: Implemented oracle target-task sampler. Geometry-valid GT OBB rows form the task pool, and seeded uniform sampling without replacement applies the manifest-defined cap. Rollout scoring later decides whether the selected crop is evaluable. No actor proposal, IoU match, visibility gate, or support gate is used.

The sampler bounds rollout cost without pre-filtering on headroom. Candidate scoring may subsequently invalidate the task when its mesh crop, current support, or rendered evidence is unusable; otherwise near-solved and negative-gain targets remain scientifically informative. A later actor-visible protocol must introduce proposal identity and observation-quality diagnostics as a separate selection stage rather than retroactively interpreting the present oracle fields as measurements.

3.2.3 `V1` Observed-Target Admission

The implemented `V1` audit path keeps this oracle sampler separate from actor-visible target selection. For each observed or predicted target record $\hat{e}$, it compares only same-class GT OBB rows using oriented 3D IoU:

$$\mu_{\text{IoU}}(\hat{e}, e) = \text{IoU}_{3\text{D}}\left(\hat{\boldsymbol{B}}_{\hat{e}}, \boldsymbol{B}_e\right)$$

$$\tau_{\text{IoU}} = 0.20$$

A GT row qualifies only when its IoU is strictly greater than 0.20. The matcher admits the observed target iff exactly one GT row qualifies:

$$n_{\text{qual}}(\hat{e}) = \text{card}(\{e \in \mathcal{E} : \kappa(\hat{y}_{\hat{e}}, y_e) = 1 \wedge \mu_{\text{IoU}}(\hat{e}, e) > \tau_{\text{IoU}}\})$$

$$a_{\text{id}}(\hat{e}) = 1 \text{ iff } n_{\text{qual}}(\hat{e}) = 1$$

Equality at 0.20 is rejected; zero qualifying rows are rejected; and multiple qualifying rows are explicitly marked ambiguous and rejected. A single compatible candidate is admitted only with IoU > 0.20. This implemented `V1` contract has no runner-up-gap or composite-μ claim. The preceding sampler description remains the current `V0` oracle-task path: its geometry-valid GT-row admission is historical `V0` evidence, not actor-visible `V1` matching.

3.2.4 Candidate View Generation

Candidate generation turns one oracle instruction into a finite action table. Every quantitative choice—source population, target cap, shell size, family weights, motion limits, pruning thresholds, rollout recipes, renderer settings, and retention policy—belongs to the resolved run manifest and report bundle. The Methods text defines semantics only; it does not designate one mutable TOML profile as canonical.

At rollout step t, candidate generation constructs a full shell

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad N_q = 60$$

with a fixed provenance component $k(i)$ per row. The core mixture contains forward-local, target-bearing, and lateral-bypass motion; the diversity challenger may additionally allocate mass to local refinement and revisit/backtrack components. The resolved manifest, rather than prose, determines which components and counts apply to a run.

For each row, the raw direction is sampled in the reference rig frame. The realistic profile uses the forward-biased Power Spherical distribution from the current implementation [DAT20]:

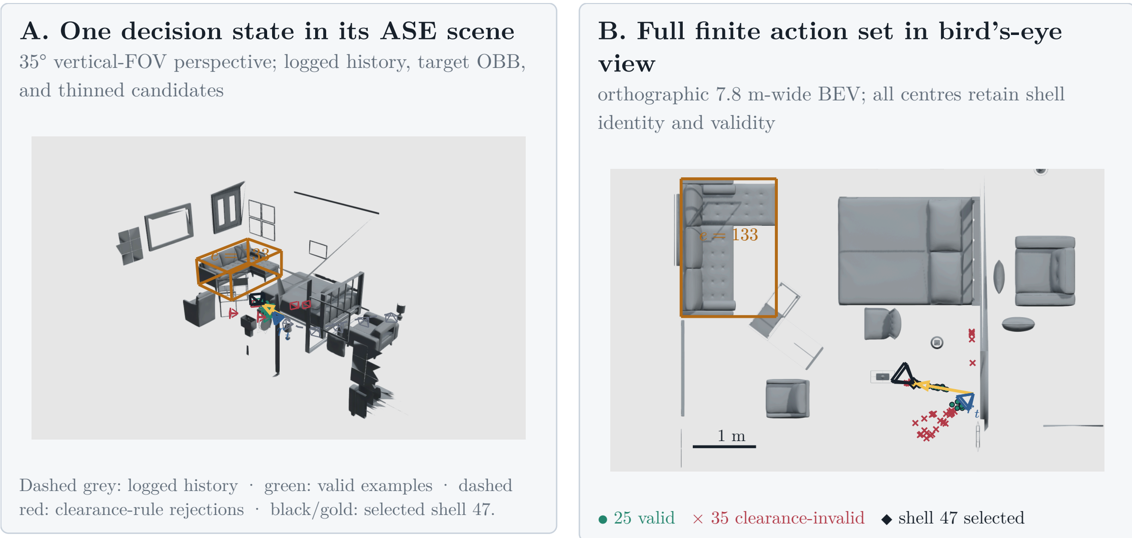
$$\boldsymbol{u}_i \sim \mathrm{PS}(\boldsymbol{e}_z, \kappa), \quad p(\boldsymbol{u}) = c_\kappa \big(1 + \boldsymbol{e}_z^T \boldsymbol{u}\big)^\kappa, \quad \kappa = 8$$

The draw is mapped into configured azimuth and elevation caps without rejection. With $\psi = \mathrm{atan2}(u_x, u_z)$ and $u_y = \sin\theta$, the cap transform is

$$\psi' = \psi \frac{\Delta_\psi}{2\pi}, \quad y' = \sin\theta_{\min} + \frac{u_y + 1}{2} \cdot (\sin\theta_{\max} - \sin\theta_{\min})$$

$$\boldsymbol{d}_i^0 = \left\| \left(\sqrt{1-y'^2}\sin\psi', y', \sqrt{1-y'^2}\cos\psi' \right) \right\|$$

The sampler draws a capped direction in the reference rig frame and reinterprets it as egocentric forward motion, target-bearing motion, lateral bypass, local refinement, or backtracking according to component provenance. Target-looking families orient their optical axis toward the oracle instruction. In this chapter that point is privileged; calling the generator target-conditioned does not make the point actor-visible.



Scene 81286, sample `ASE_81286_Atek_000035`, rollout row 73, step row 121 24 forward-local + 24 target-bearing
1. 12 lateral-bypass candidates. Dense mesh layers are
z-buffered raster; OBBs, paths, centres, and wire frusta remain vector geometry.

Figure 3: One pinned finite-candidate decision state from ASE scene 81286, sample `ASE_81286_Atek_000035`, rollout row 73, and step row 121. Panel A uses a 35-degree vertical-FOV perspective view to place logged camera history, root state, target OBB, selected path, and a deterministically thinned set of wire frusta in the real scene. Panel B uses a 7.8-metre-wide orthographic bird's-eye view and retains all 60 candidate centres: 25 rows are admissible, 35 are hard-rejected by the clearance rule, and oracle-greedy selects shell 47. Dense scene geometry is z-buffered; OBBs, paths, centres, and camera glyphs remain vector overlays. Full eye, look-at, up, clipping, and resolution parameters are recorded in the figure JSON. This is an auditable contract example, not a policy-performance result.

The three core position families then reinterpret this capped direction. Let $\boldsymbol{f} = \boldsymbol{e}_z$ be the rig-forward unit vector, $\boldsymbol{b}_e$ the supplied target bearing in the reference frame, $\boldsymbol{l}_e = \|\boldsymbol{e}_y \times \boldsymbol{b}_e\|$ the horizontal lateral direction, and $\boldsymbol{e}_y$ the world-up direction expressed in the sampling frame. The family directions are:

$$\boldsymbol{d}_i^{\text{forward}} = \|\boldsymbol{f} + \alpha_f(\boldsymbol{d}_i^0 - (\boldsymbol{d}_i^0 \cdot \boldsymbol{f})\boldsymbol{f})\|, \quad \alpha_f = 0.45$$

$$\boldsymbol{d}_i^{\text{target}} = \|\boldsymbol{b}_e + \alpha_t(\boldsymbol{d}_i^0 - (\boldsymbol{d}_i^0 \cdot \boldsymbol{b}_e)\boldsymbol{b}_e)\|, \quad \alpha_t = 0.4$$

$$\boldsymbol{d}_i^{\text{bypass}} = \left\|0.55\boldsymbol{b}_e + 0.85\,\operatorname{sign}\!\left(d_{i,x}^0\right)\boldsymbol{l}_e + \operatorname{clip}\!\left(d_{i,y}^0, -0.35, 0.35\right)\boldsymbol{e}_y\right\|$$

Finally, the sampler draws a radius and transforms the reference-frame offset into world coordinates:

$$r_i \sim \mathcal{U}(0.25, 1.1), \quad \boldsymbol{c}_i^w = \boldsymbol{T}_r^w\left(r_i \boldsymbol{d}_i^{k(i)}\right)$$

The forward-local family keeps the reference rig orientation. Target-looking families orient the camera toward the supplied target center $\boldsymbol{p}_e$:

$$\boldsymbol{z}_i^w = \|\boldsymbol{p}_e - \boldsymbol{c}_i^w\|, \quad \boldsymbol{y}_i^w = \left\|\boldsymbol{e}_y - \left(\boldsymbol{e}_y^T \boldsymbol{z}_i^w\right)\boldsymbol{z}_i^w\right\|, \quad \boldsymbol{x}_i^w = \boldsymbol{y}_i^w \times \boldsymbol{z}_i^w$$

These equations describe the sampler, not an optimal proposal distribution. The forward-local family preserves egocentric continuity, the target-bearing-local family moves along the target ray, and the lateral-target-bypass family introduces side-step views; optional challenger families test smaller corrections and reversals. Candidate-profile utility must be judged after pruning. A store in which target-aware families rarely survive cannot support a target-conditioned planning claim.

Pruning converts the full shell into a compact valid-action table. A row remains valid only if it lies in the snippet occupancy support, stays clear of the GT mesh, avoids straight-line path collision, and satisfies local egocentric motion limits:

$$\|\boldsymbol{o}_i\|_2 \leq 1.0 \text{ m}, \quad |\Delta h_i| \leq 0.25 \text{ m}, \quad \max\!\left(0, -o_{i,z}\right) \leq 0.25 \text{ m}, \quad \Delta\psi_i \leq 70 \text{ deg}$$

The full shell is retained with position, strategy, mixture, sampling probability, rule masks, diagnostics, and invalid-reason bitsets. Panel B of Figure 3 makes the distinction concrete for one stored table: invalid rows remain inspectable but sit outside the admissible set. Invalid candidates cannot enter Q_H selection, stochastic normalization, or loss targets. Conversely, a feasible row with weak target support or low expected gain remains a valid low-utility example rather than receiving an invalid-reason code. A manifest-defined root-support threshold may reject an entire rollout task when too few actions remain:

$$N_{\text{valid}} \geq \max\big(12, \text{ceil}\big(0.25N_q\big)\big)$$

This threshold is a data-support guard, not a reward threshold. Preflight reporting must retain rejected-root counts and failure reasons by candidate family.

3.2.5 Rollout Branch Sampling and Dataset Impact

Rollout recipes select and retain finite chains from the valid action table. The implemented families are uniform valid sampling, one-step oracle greedy selection, bounded oracle lookahead, and temperature-softmax sampling. Their horizon, branch factor, beam width, temperature, and seed are resolved parameters. These recipes generate replay diversity and bounded references; they do not constitute a learned policy.

Bounded oracle lookahead can select a different first action from one-step greedy because it ranks retained finite-horizon chains rather than immediate gain alone (Figure 4). The persisted artifact contains these selected or beam-retained chains and their full per-step candidate shells; it is not an exhaustive materialization of the counterfactual action tree.

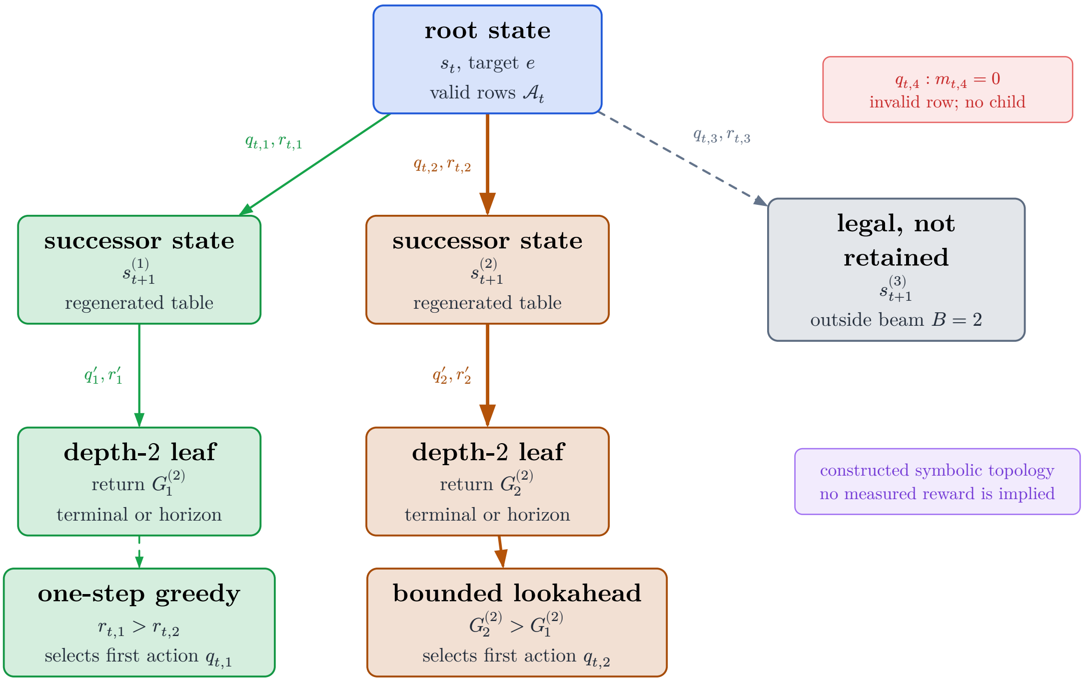


Figure 4: Constructed symbolic topology of the bounded oracle-lookahead reference. Nodes are counterfactual states and edge labels are candidate actions with immediate rewards. Only valid first-action rows produce children; the beam retains a bounded prefix set, and invalid rows receive no branch. The inequalities illustrate how the selected first action can differ from one-step greedy without inventing measured reward values.

For stochastic branches, let s_i be the finite oracle score of valid row i. The robust logit used for temperature-softmax is

$$\ell_i = \frac{s_i - \mathrm{median}(s)}{\mathrm{IQR}(s)\tau}, \quad P(i \mid m_i = 1) = \frac{\exp(\ell_i)}{\sum_{j:m_j=1}} \exp(\ell_j)$$

The target source defines the supervised task, the candidate mixture defines learnable action support, validity defines admissibility, and the recipe defines which chains enter replay. Reporting must therefore cover target-task coverage, valid fanout and invalid reasons by family, selected-family diversity, gain distributions, and retention cost before attributing policy behavior to non-myopic planning. The paired train-only pilots are bandwidth and candidate-profile probes; they are non-confirmatory and provide no held-out policy-performance evidence.

3.2.6 Reconstruction-Quality Metric

Let $C_e(\mathcal{P}_t)$ denote the oracle-only crop of accumulated evaluation points to the selected target region. In the thesis protocol, its root points are reconstructed from the observed prefix of ASE GT depth; a candidate contributes renderer-derived depth that is backprojected and fused with the same root cloud. MPS semi-dense points remain actor-visible representation input and a legacy diagnostic source, not the current oracle evaluation cloud.

Writing $P = C_e(\mathcal{P}_t)$ and $M = \mathcal{M}_e^{\mathrm{GT}}$ for the target-cropped point set and mesh, the selected directional accuracy is the mean squared distance from each reconstructed point to its closest GT triangle,

$$D_{P\to M}\big(\mathcal{P}, \mathcal{M}^{\mathrm{GT}}\big) = \frac{1}{\|\mathcal{P}\|}\sum_{\boldsymbol{p}\in\mathcal{P}} \min_{\boldsymbol{f}\in\mathcal{F}^{\mathrm{GT}}} d(\boldsymbol{p}, \boldsymbol{f})^2$$

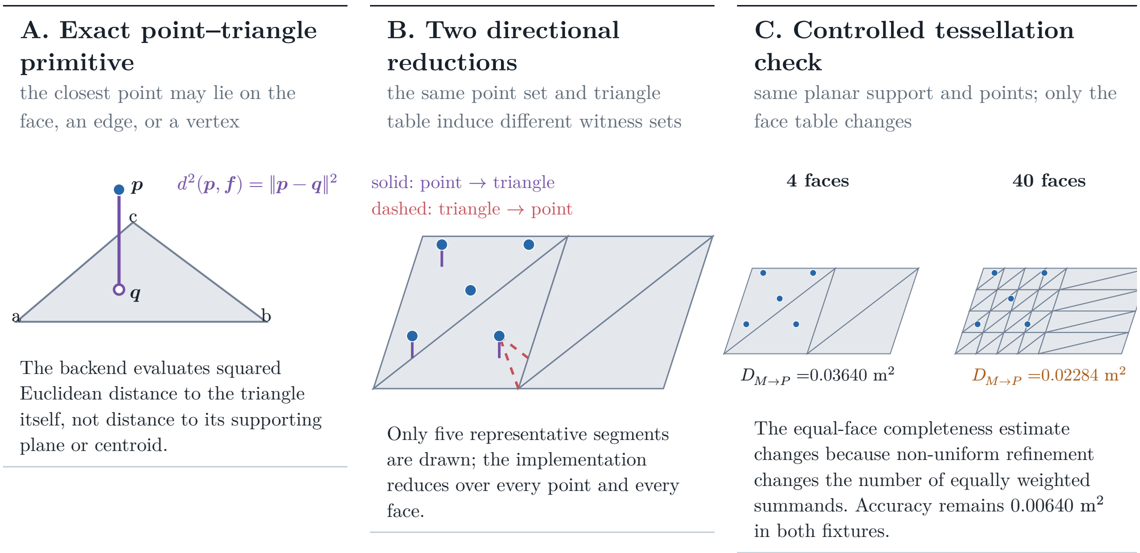
It penalizes reconstructed points that lie away from the target surface and therefore exposes extra, noisy, or misregistered geometry. The reverse term is the mean squared distance from each GT face to its closest reconstructed point,

$$D_{M\to P}\big(\mathcal{P}, \mathcal{M}^{\mathrm{GT}}\big) = \frac{1}{\|\mathcal{F}^{\mathrm{GT}}\|}\sum_{\boldsymbol{f}\in\mathcal{F}^{\mathrm{GT}}} \min_{\boldsymbol{p}\in\mathcal{P}} d(\boldsymbol{p}, \boldsymbol{f})^2$$

It penalizes target faces without nearby reconstructed evidence and therefore exposes missing support. Their sum is the bidirectional point–mesh error,

$$\Delta_t^e = d\big(C_e(\mathcal{P}_t), \mathcal{M}_e^{\mathrm{GT}}\big) = D^e_{P\to M,t} + D^e_{M\to P,t}$$

whose directional terms and total have units of square metres. The current and candidate-augmented clouds pass through the same deterministic fusion and point-cap configuration before scoring. This fusion step makes the before/after comparison operationally reproducible; it does not turn the point set into a continuous surface measure.



Controlled validity fixture computed by the repository's PyTorch3D metric and exact Trimesh closest-point witnesses; it is not an ASE performance result.

Figure 5: Point–mesh metric contract and controlled validity fixture. Panel A isolates the exact point-to-triangle primitive. Panel B distinguishes the point-to-face accuracy and face-to-point completeness reductions using computed closest-point witnesses. Panel C holds the planar support and point set fixed while changing only the triangle table; the measured equal-face completeness term changes from 0.03640 to 0.02284 m^2. These values are generated by the repository's PyTorch3D metric on a synthetic fixture and demonstrate a tessellation-sensitivity mechanism; they are not an ASE performance result.

The completeness reduction gives every mesh face equal weight. It is deliberately neither an area-weighted surface integral nor invariant to retessellation: subdividing part of an unchanged surface changes how much that region contributes. The controlled fixture in Figure 5 demonstrates this property. It bounds the interpretation of the operational estimand but does not invalidate comparisons that share one fixed mesh and scoring contract.

Metric family	Question answered	Dependency and thesis role
Equal-face bidirectional point–mesh error	Are reconstructed points close to GT faces, and does every GT face have nearby point evidence?	Depends on point fusion and mesh tessellation; selected operational RRI error.
Point-sampled Chamfer / accuracy–completeness	Are two sampled surface point sets mutually close?	Depends on sampling density, randomness, norm, and squaring convention; important alternative, not the current metric.
Thresholded precision, recall, and F-score	What fraction of predicted and reference samples fall within tolerance τ, and how do the two fractions balance?	Interpretable at a physical tolerance but threshold-dependent; useful diagnostic rather than the scalar RRI owner.
TSDF, occupancy, coverage, or information gain	How much volumetric surface, free space, or previously unknown space is reconstructed or observed?	Depends on grid, truncation, and visibility definitions; answers a different question from target surface quality.

Table 2: Reconstruction-metric alternatives and their role in this thesis. No alternative is promoted to a co-primary metric.

For comparison, a common squared point-sampled Chamfer convention draws a reference set Q_e from the mesh and averages nearest-neighbour distances in both directions,

$$D_{\text{PS-Chamfer}(\mathcal{P},\mathcal{Q})} = \frac{1}{\|\mathcal{P}\|}\sum_{\boldsymbol{p}\in\mathcal{P}} \min_{\boldsymbol{q}\in\mathcal{Q}} \|\boldsymbol{p}-\boldsymbol{q}\|_2^2 + \frac{1}{\|\mathcal{Q}\|}\sum_{\boldsymbol{q}\in\mathcal{Q}} \min_{\boldsymbol{p}\in\mathcal{P}} \|\boldsymbol{q}-\boldsymbol{p}\|_2^2$$

Its apparent symmetry does not remove sampling-density dependence: changing the number or distribution of samples changes the finite approximation. Some benchmarks also use unsquared distances, so the norm and squaring convention are part of the metric contract rather than an interchangeable naming detail. At tolerance τ, precision counts reconstructed samples with a reference neighbour closer than τ, recall counts reference samples with a reconstructed neighbour closer than τ, and $F_{1,\tau} = 2\text{Prec}_\tau \frac{\text{Rec}_\tau}{\text{Prec}_\tau+\text{Rec}_\tau}$. These scores are physically interpretable but can change with the selected threshold and discard error magnitude on either side of it. Volumetric coverage and information gain are valuable for exploration and mapping, yet they measure discovered space or uncertainty reduction rather than the target reconstruction-quality change defined here.

The implementation crops mesh faces when any vertex lies inside the oriented target OBB and evaluates the selected point–mesh scorer. Geometry-invalid candidates are removed by the hard action mask and retain persisted candidate reason codes. Empty mesh crops, insufficient current support, or unusable renders instead invalidate the separate oracle evaluation: the recipe either skips the affected row or table, or clears oracle-label validity and Q-training eligibility, while reporting the oracle failure independently rather than assigning a candidate reason code.

This thesis therefore treats the metric as an operational estimand over a frozen evaluation pipeline, not as a representation-independent surface quantity. Every compared policy must share the same target mesh and crop, ASE-depth source, renderer and backprojection, fusion and point cap, and metric configuration. A future real-data evaluation that replaces ASE depth with MPS semi-dense evidence must separately quantify its texture-, gradient-, and uncertainty-dependent sampling effects; those concerns are not properties of the current oracle cloud.

3.2.7 Target-Specific RRI

The thesis distinguishes three related quantities rather than treating them as aliases: state-relative RRI is a current-state diagnostic, target-root gain is the additive learning reward, and endpoint gain is the fixed-budget evaluation metric. All are normalized reductions of the same target-cropped error, but their denominators and roles differ.

The state-relative marginal target RRI remains the VIN-compatible candidate diagnostic:

$$\mathrm{RRI}^{e}_{t,i} = \frac{\Delta^{e}_{t} - \Delta^{e}_{t|i}}{\max(\Delta^{e}_{t}, \varepsilon)}$$

The implementation also reports its selected-chain running sum:

$$C_t^{\mathrm{RRI},e} = \sum_{k=0}^{t-1} \mathrm{RRI}_{k,a_k}^{e}$$

Because the marginal RRI denominator changes with state, this cumulative diagnostic does not telescope to endpoint improvement. The immediate training reward instead adapts the same error reduction to a target crop and normalizes by the rollout-root target error [Fra+25]:

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

Under matched sequential states and unit discount, the shared cumulative-root equation records the additive reward sum as a root-normalized endpoint difference:

$$J_t^e = \sum_{k=0}^{t-1} r_k^e = \frac{\Delta_0^e - \Delta_t^e}{\max(\Delta_0^e, \varepsilon)}$$

The finite-horizon return is the discounted sum of those target rewards along a selected counterfactual branch. It is a training target for Q_H, not a claim that the deployed system has an online continuous-control policy:

$$G_{t,e}^{(h)} = \sum_{k=0}^{\min(h,b_t)-1} \gamma^k r_{t+k}^e$$

Endpoint gain is the primary fixed-budget comparison metric because it measures the target quality after the same number of acquisitions for each policy:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

The log-gain variant is retained only as an internal scale-sensitivity ablation:

$$J_{e,\log}^{(H)} = \log(\Delta_0^e + \varepsilon) - \log(\Delta_H^e + \varepsilon)$$

$J_e^{(H)}$ is the fixed-budget evaluation metric, $G_t^{(H)}$ is the learning return, and $J_{e,\log}^{(H)}$ is a sensitivity diagnostic. The endpoint equation uses $\Delta_0^e + \varepsilon$, whereas the additive root-gain equation uses $\max(\Delta_0^e, \varepsilon)$; the two canonical metrics are therefore not exactly equivalent, even when horizon, geometry, and acquisition budget match. State-relative one-step RRI remains a VIN-compatible diagnostic. The resolved manifest must freeze the discount, clipping, target cap, crop policy, and all evaluation-geometry parameters for each reported experiment.

3.3 Replay Stores and Diagnostic Evidence

Remove or rewrite before submission: Move schema names, directory names, joins, codecs, and DTO-level mechanics that do not change the scientific protocol to the implementation appendix. Keep lineage, missingness, leakage prevention, and reproducibility guarantees in the main text.

Source: this section; aria_nbv/aria_nbv/rollouts/zarr_store.py

Gate: every remaining implementation term is necessary to reproduce or interpret the experiment

The pipeline materializes two stores with different ownership. Immutable `vin_offline/` caches logged snippet evidence and expensive one-step oracle products. Standalone `rollouts.zarr/` references those source rows and stores target tasks, retained rollout chains, per-step finite candidate shells, masks, reason codes, selected-action successor evidence, and a derived Q_H view. This separation prevents counterfactual experiments from mutating the source substrate and prevents one-step candidate labels from being mistaken for multi-step replay.

The source store is immutable because it defines the logged actor substrate. Its manifest records source identity, split, schema, materialized blocks, and provenance. A source row may contain frozen EVL/VIN fields, one-step candidate poses and labels, compact rendering payloads, and semidense geometry/history. These fields support scorer training and rollout construction, but they do not themselves define target-conditioned finite-horizon data. Exact directory names, array groups, chunking, and codec choices are versioned reproducibility metadata rather than scientific entities, so they are not reproduced as directory-tree figures in the main text.

The rollout store is normalized around replay identity. One source can produce multiple target rows, each target can produce multiple recipe and retained-chain rows, each chain expands into selected steps, and each step owns a full candidate shell. The store therefore captures selected or beam-retained chain evidence, not the complete counterfactual search tree. Padded `q_h/` arrays are a validated cache over the factual row tables. Invalid candidate rows remain inspectable, but action, training, and bootstrap masks exclude them.

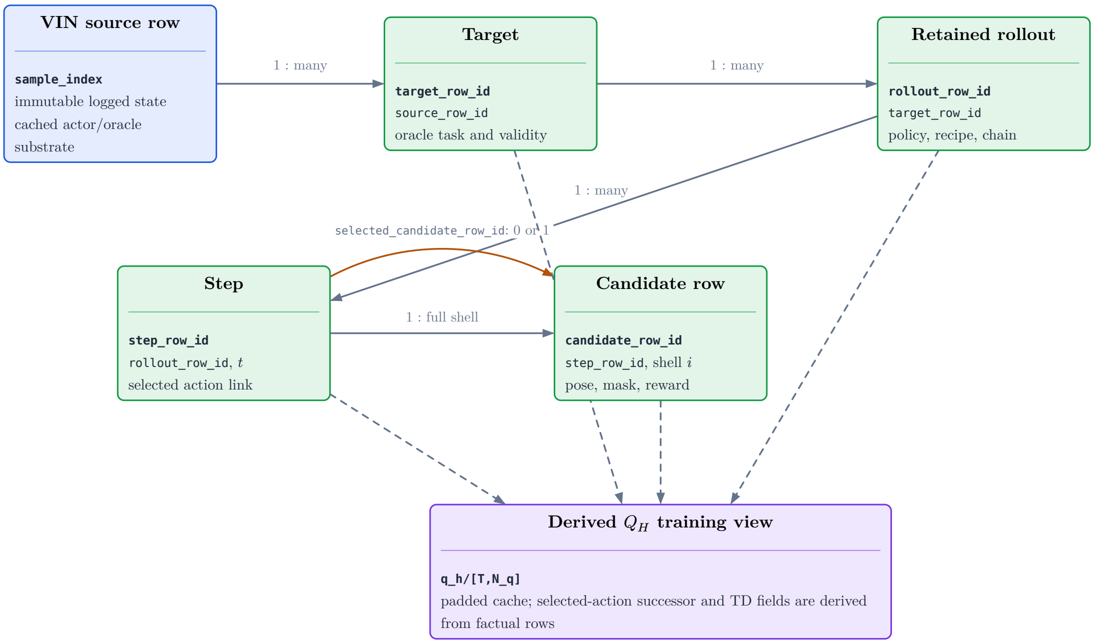


Figure 6: Normalized lineage of the implemented replay evidence. An immutable VIN source row may define several oracle target tasks; each target may produce several retained policy chains; each chain contains ordered steps; and each step owns one full candidate shell. The step row may identify one chosen candidate through `selected_candidate_row_id`; selected-action successor and TD fields are then constructed in the derived `q_h/` join rather than persisted as a separate transition table.

Evidence family	Interpretation contract
Sources and targets	Manifest-backed task coverage; not proof of actor-visible target discovery.
Candidates and invalidity	Full-shell support with separate hard-action, training, padding, and future deployable-feasibility roles.
Retained chains and steps	Recipe-selected evidence; not a persisted exhaustive search tree.
Selected depth	Chosen-action successor observation with calibration and source role; actor input only under an explicitly admitted later-state protocol.
Q_H view	Derived training cache whose rewards and masks must agree with factual rows; not a scene-memory representation.

Table 3: Interpretation contract for rollout-store audits. Numeric values are rendered from the resolved report bundle in the experiment and reproducibility sections.

Selected-depth persistence stores only the depth raster and calibration for the chosen action at each retained step. It is sufficient to reconstruct the selected-observation prefix without duplicating dense all-candidate renders, but persistence does not decide visibility. A `CF-GT` reader may use previously selected GT-mesh depths to build a privileged dynamic state; a deployable reader must instead consume a declared sensor-like or observed source. The current unselected candidate renders remain oracle-only in every student protocol. Selected depth is also not an independently scored endpoint artifact.

Likewise, rollout rows summarize final cumulative selected-chain metrics; they do not preserve every rejected branch or a policy-neutral endpoint reconstruction. These limitations must be resolved by matched endpoint re-evaluation before confirmatory policy comparison.

Scientific reporting is generated from the same inspection frames used by the diagnostic application. The main text needs target-task coverage, candidate validity and invalid reasons, family survival and selection, selected-history sanity, gain distributions, source-role counts, and runtime/storage summaries. Exact schema columns, compression, chunking, hashes, and cluster invocation belong in the reproducibility appendix and must be read from the resolved manifest and report bundle. Development bandwidth pilots are train-only feasibility checks; their counts and throughput may size later jobs but cannot support held-out reconstruction or policy claims.

4 Method

ARIA-NBV is a target-conditioned selection problem over a finite candidate table. The current method path records an implemented replay, read-model, DTO, profile, and return substrate. Its evidence boundary ends before a production scorer or learned policy: no checkpoint or held-out policy result is claimed here.

The method fixes the actor/oracle boundary, the finite replay contract, the three mask meanings, the frame and support limits, and the metric semantics before any policy claim. The following sections record these implemented contracts; development-mode material retains the future scorer design and its evaluation gates separately.

The planned scorer direction is a bounded fixed-horizon value model. Remaining budget is part of the actor state, and the candidate table is the finite support on which selection, training, and bootstrap are defined. Requested-horizon queries, exact-$H = 2$ certification, recursive fitted values, Monte Carlo returns, CQL/BCQ controls, alternative encoders, and alternative policies are development evidence or future comparisons, not the accepted flow.

4.1 Actor State and Representation Boundary

4.1.1 Implemented actor boundary

The implemented actor DTO carries root evidence, target fields, root-relative candidate poses, factual selected-pose history, remaining budget, candidate support, and actor-validity masks. It does not carry rewards, target-RRI labels, or audit lineage. The `qh_cf0_v1` profile has no selected counterfactual depth. `qh_cfplus_gt_depth_v1` is a named privileged selected-depth profile and is rejected for deployable construction unless privilege is explicit. These profiles are not interchangeable evidence cohorts.

The actor/oracle separation is strict. GT meshes, target crops, current all-candidate renders, associations, target gains, state-relative RRI diagnostics, and returns are supervision, evaluation, or audit fields. GT target geometry may define the current oracle-task instruction, but it is not an ordinary actor observation.

Previously selected GT depth can enter only the explicitly privileged CF+ protocol; it must not be described as the actor-visible CF0 path.

The implemented carrier is an information boundary, not a production scorer. In particular, the DTO does not prove task-sufficient dynamic reconstruction memory, scorer invariance, or policy utility.

4.1.2 Planned actor state and support limits

The planned production state for target e, candidate $q_{t,i}$, and step t is expressed with existing shared notation:

$$\boldsymbol{h}_{t,e,i} = \mathrm{Read}\big(\boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{h}_e^{\mathrm{tgt}}, q_{t,i}, \boldsymbol{H}_t, t, H\big)$$

The fixed-horizon path uses remaining budget b_t as time-to-go context. A requested residual horizon is not a second primary interface. The actor state may combine immutable root evidence, causal selected-observation memory, target context, candidate-local queries, and ordered history, but every field needs a declared source and availability mask. No future observation or unselected candidate render may enter the state.

EFM3D/EVL support is local. Its persisted voxel pose and finite extent define where lifted features are supported; outside that extent is missing evidence, not an ordinary zero feature and not automatically an invalid action. The planned state may add semidense or fused points and sparse ray-aware support, but no task-sufficient dynamic memory is currently available. The state remains bounded by the logged EVL/support fields and by the finite candidate and replay support.

Carrier	Role in the primary path
root semidense/EVL evidence	bounded actor context with explicit support metadata
causal selected-observation memory	not part of this substrate; requires source and leakage tests
GT mesh and rendered depth	oracle label/evaluation evidence, never ordinary actor input

Table 4: Scene-carrier design space and information roles.

An origin or yaw convention must not become a shortcut. Relative geometry is an adapter/DTO contract: world-frame facts are retained for provenance, while candidate features are derived in the admitted root or local frame. This does not claim scorer-level exact SE(3) invariance; gravity, scale, target extent, camera direction, occlusion, and temporal order remain task variables.

Protocol	Selected evidence	Scientific role
`qh_cf0_v1`	root actor evidence; no selected CF-GT depth	actor-safe protocol; deployable only with an observed target protocol
`qh_cfplus_gt_depth_v1`	root evidence plus selected GT depth	explicit privileged upper-bound/control cohort; never ordinary actor evidence
Oracle	GT mesh, target crop, rendered candidates and labels	label generation and re-evaluation only

Table 5: Q_H information protocols and their roles.

4.1.3 Development and future representation evidence

The planned carrier ladder starts with root semidense/EVL evidence, then tests selected-observation point or ray memory only when a diagnosed support failure requires it. Target-centred re-lifting, point appearance descriptors, sparse encoders, object-aware renderable memory, and global scene-language carriers are alternatives. They remain outside the primary method until matched support, leakage, runtime, and held-out target-RRI evidence exists.

Alternative	Promotion condition
selected point/ray memory	causal fusion, source masks, and target-conditioned held-out benefit
target-centred EVL or appearance carrier	logged-source visibility and matched support ablation
sparse, equivariant, or renderable encoder	simpler carrier failure plus fair architecture control

Table 6: Development-only representation alternatives.

4.2 Descriptor and Encoding Protocol

4.2.1 Implemented replay and DTO projection

The replay/read-model projection preserves source, target, rollout, step, candidate, diagnostic, and lineage identities. Its padded `q_h/` view is a storage/read contract; padding does not create factual transitions or actor observations. The actor projection and supervision projection remain separate even when they are collated in one batch.

No production scorer consumes this projection. The current DTO/read-model boundary keeps root context, candidate rows, target context, factual history, remaining budget, and support masks subject to their admitted protocol; `q_train_mask`, target-root-gain labels, target-RRI diagnostics, selected transitions, and lineage remain supervision or audit rather than scorer features.

The actor-facing target descriptor is `#symb.entity.target_desc`; its source and availability must be explicit. The current oracle task uses GT-derived target geometry, so a deployable target-conditioned scorer requires an observed or predicted target protocol before policy evidence can be claimed.

4.2.2 Frame and encoding contract

The implemented frame discipline is at the DTO/adapter boundary: root and candidate poses have declared frames and the history mask is true only for factual selected poses preceding the query state. World-frame values remain audit facts. This contract does not establish scorer invariance or prove that any encoder is task-sufficient.

The current boundary does not include a production candidate encoder. It preserves declared frames, separates padding from modality absence, and keeps candidate-family and sampler provenance audit-only.

4.2.3 Development alternatives

The planned scorer input follows the shared contract:

$$\mathcal{I}_{t,e} = \left(\boldsymbol{h}_e^{\mathrm{tgt}}, \boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{H}_t, \boldsymbol{b}_t, t, H, \left\{\boldsymbol{x}_{t,i}, \boldsymbol{e}_{a|i}^{\mathrm{rel}}, m_{t,i}, \boldsymbol{\rho}_{t,i}\right\}_{i=1}^{N_q}\right)$$

The planned candidate representation uses shared equations for pose and target relations:

$$\boldsymbol{T}_{r_t,i}^{\mathrm{rel}} = \boldsymbol{T}_{w,r_t}^{-1}\boldsymbol{T}_{w,c_{t,i}}, \quad \boldsymbol{\delta}_{r_t,i}^{p} = \boldsymbol{R}_{r_t}^{\top}\left(\boldsymbol{c}_{t,i} - \boldsymbol{c}_{r_t}\right), \quad \boldsymbol{R}_{r_t,i} = \boldsymbol{R}_{r_t}^{\top}\boldsymbol{R}_{t,i}$$

$$\boldsymbol{h}_{t,i}^{\mathrm{pose}}(q_{t,i}; r_t) = \mathrm{concat}\left(\boldsymbol{\delta}_{r_t,i}^{p}, \mathrm{R6D}\left(\boldsymbol{R}_{r_t,i}\right), \left\|\boldsymbol{\delta}_{r_t,i}^{p}\right\|_2, \mathrm{atan2}\left(\delta_{r_t,i}^{y}, \delta_{r_t,i}^{x}\right), \Delta h_{t,i}, \boldsymbol{u}_{t,i}^{\mathrm{up/frustum}}\right)$$

Missing EVL, appearance, ray, and target fields require availability/source masks. Padding is separate from modality absence. Candidate-family and sampler provenance remain audit-only unless a named ablation promotes them.

Requested-horizon embeddings, separate $Q_1, ..., Q_H$ heads, learned point or ray encoders, spherical-harmonic features, target-centred re-lifting, DINO-on-point, and other carrier choices are alternatives. They may be compared only after the fixed-H scorer boundary and source protocol are frozen; none is an implemented primary encoding.

4.3 Finite Candidate and Replay Contract

4.3.1 Implemented finite action and replay substrate

At step t, candidate generation returns a finite table $\mathcal{Q}_t$ with stable row identity, hard validity, and versioned invalid reasons:

$$\mathcal{Q}_t = \left\{q_{t,i}\right\}_{i=1}^{N_q}, \quad \mathcal{A}_t = \left\{i \in \left\{1, ..., N_q\right\} : m_{t,i} = 1\right\}$$

Three masks remain separate: the valid-action mask selects physically admitted rows; `q_train_mask` is stricter and additionally requires finite oracle labels; padding marks absent rows in the materialized rectangular view. Invalid rows cannot enter selection, loss, or bootstrap. They remain available for diagnostics. An all-invalid successor has no utility value: it sets no-bootstrap support explicitly rather than becoming a zero or an imputed target.

The implemented replay transition is:

$$(x_{t+1}, \boldsymbol{H}_{t+1}, b_{t+1}, \mathcal{Q}_{t+1}) = \mathrm{Step}\left(x_t, \boldsymbol{H}_t, b_t, q_{t,a_t}, \xi_t\right)$$

It updates pose, factual selected-pose history, remaining budget, lineage, and regenerated candidate support. It does not imply a new RGB observation, a recomputed EVL field, or a fused selected-depth scene state. A pose/history chain is therefore an `S0-pose` replay state, not a task-sufficient visual successor.

4.3.2 Selected observations and oracle roles

Selected depth is a typed observation with calibration, validity, pose, and source role. The `CF-GT` source is privileged and belongs only to `qh_cfplus_gt_depth_v1`. It is not an ordinary actor input and cannot be silently mixed with `qh_cf0_v1`. Unselected candidate renders never enter the actor state.

The current replay carrier does not update a visual successor state. Missing source or modality evidence is therefore not replaced with a fabricated descriptor; selected depth remains typed metadata and is admitted to actor tensors only under the explicit privileged CF+ protocol.

4.3.3 Development and future evidence

The planned causal update is expressed by the shared equation:

$$\boldsymbol{M}_{t+1}^{\text{ray}} = \text{Fuse}\big(\boldsymbol{M}_t^{\text{ray}}, \boldsymbol{P}_{t+1}^{\text{selected-obs}}(a_t), \mathcal{R}_{t+1}^{\text{selected-obs}}(a_t)\big), \quad a_t \in \mathcal{A}_t$$

Only selected evidence may update a future successor state. Missing source or modality evidence is represented by masks, not fabricated descriptors.

Exact $H = 2$ targets, recursive targets beyond the supported chain, requested-horizon queries, and Monte Carlo behavior returns are separate development controls. They require their own admitted transition/support evidence and cannot turn sparse replay into dense long-horizon support.

4.4 Geometric and Mask Acceptance Tests

4.4.1 Acceptance boundary

The implemented acceptance boundary is the DTO, replay, and fitted-Q adapter contract. It is not a scorer-level architecture or invariance result. A production scorer must be admitted only after it preserves the row, source, frame, and mask contracts.

The required tests are literal. Permuting candidate rows must permute row-aligned outputs; changing invalid-row contents must not change valid-row values; padding must not become an action; `q_train_mask` must not widen valid action support; and an all-invalid successor must not be bootstrapped. Frame-transform tests apply to the DTO/adapter representation. They do not authorize a claim of exact SE(3) scorer invariance.

Actor/oracle tests must also reject GT target gains, GT associations, mesh distances, candidate renders, and target crops in actor inputs. A selected-depth source is accepted only under the explicit privileged CF+ profile and matching geometry contract. `qh_cf0_v1` and `qh_cfplus_gt_depth_v1` are separate experimental cohorts, not interchangeable checkpoints.

4.4.2 Primary architecture direction

The planned scorer is a bounded fixed-H candidate-value model. It reads actor-admitted root/context features, target context, factual history, remaining budget, and a finite candidate table, then emits one continuous value per candidate row. Masked selection uses only valid candidate rows. The current substrate provides the injected-scorer seam and owns transition validation, loss, target synchronization, and contract checks; scorer construction and policy evaluation remain a future deliverable.

Primary component	Contract
fixed-H candidate-value scorer	one continuous value per finite candidate row, remaining budget in state, valid-action masked selection
injected-scorer adapter	selected-transition loss, target synchronization, profile and contract validation

Table 7: Primary geometric-learning contract.

The fixed-H boundary keeps candidate interaction and representation choices subordinate to the information and support contracts. It does not require candidate-to-candidate attention, recurrent memory, or exact equivariant layers. Those choices may be evaluated only after a baseline scorer passes row, mask, source, frame, and support tests.

The development ladder is A0 independent row scorer, A1 candidate-to-state query, A2 set context, A3 masked candidate interaction, and later temporal or equivariant variants. Requested-horizon conditioning is an alternative time-query design, not a primary architecture. A level advances only on a diagnosed failure and matched held-out evidence under the same actor protocol.

4.5 Target-Conditioned Finite-Horizon Value Model

4.5.1 Implemented learning substrate

The implemented component is scorer-independent fitted-Q infrastructure: an injected scorer receives actor DTOs, while the Lightning adapter owns admitted-row loss, target synchronization, and selected-transition backups. This is an implemented substrate contract, not a production finite-horizon scorer, checkpoint, or policy result.

The persisted reward semantics are additive target-root gain. The canonical one-step reward is:

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

and the finite-horizon return is:

$$G_{t,e}^{(h)} = \sum_{k=0}^{\min(h,b_t)-1} \gamma^k r_{t+k}^e$$

Target-root gain telescopes under matched selected states and $\gamma = 1$:

$$J_t^e = \sum_{k=0}^{t-1} r_k^e = \frac{\Delta_0^e - \Delta_t^e}{\max(\Delta_0^e, \varepsilon)}$$

State-relative target RRI is a separate diagnostic for current-state improvement. It is not an alias for `target_root_gain` and is not additive.

No production scorer is evaluated here. Any future scorer would remain bounded by the generated candidate support, hard-validity regime, represented actor state, and offline transition support; the current DTO frame and profile contracts do not prove task-sufficient memory, scorer invariance, or policy optimality.

4.5.2 Development scorer direction

The planned scorer is a bounded fixed-horizon value model with remaining budget in state. For finite candidate support, its intended value and masked decision rule use the shared equations:

$$Q_{h,e}(s_t, i) = \mathbb{E}\Big[G_{t,e}^{(h)} \mid s_t, a_t = i\Big], \quad i \in \mathcal{A}_t, \quad 1 \le h \le b_t, \quad Q_{0,e}(s, i) = 0$$

$$a_t^\theta = \operatorname*{argmax}_{i:m_{t,i}=1} Q_{h,\theta,e}(s_t, i)$$

A requested-horizon model, exact-$H = 2$ target, recursive targets, Double-Q selector/evaluator variants, Monte Carlo behavior returns, CQL/BCQ support controls, residual heads, and alternative encoders are development evidence. They remain separately identified by horizon, state/source protocol, support, and estimand. CQL/BCQ are not part of the primary flow and are introduced only if unsupported-action diagnostics justify an offline-conservatism study.

Exact-$H = 2$ is a certification control, not the method definition. A behavior return estimates Q^μ, not the greedy finite-support value $Q^\star$. No alternative enters the accepted primary path until it has an admitted scorer, compatible checkpoint, and matched oracle re-evaluation.

5 Experimental Design

The experiments separate feasibility evidence from policy evidence. Feasibility requires a frozen scene-level population, reproducible target and candidate support, valid replay linkage, and measured oracle throughput. Policy inference additionally requires a held-out scene split, matched acquisition budgets, independent endpoint oracle evaluation, and an implemented actor-visible decision rule. The current train-only bandwidth pilots address only the first category: incomplete runs and resource failures characterize the generation system, but they cannot support comparisons between policies.

5.1 Study Population and Evidence Gates

The study population is split by scene into train, validation, and held-out test manifests. The implemented generator samples oracle target tasks from geometry-valid GT rows; this establishes oracle-task coverage, not observed-target matching or deployable target support. GT target geometry, candidate renders, and target-RRI labels remain oracle assets.

5.1.1 Current generator evidence

The current data generator constructs its task pool from non-padded GT OBB rows, retains only finite positive geometry for first-pass eligibility, and applies a seeded uniform cap (three tasks per snippet by default). The writer consumes the selected oracle rows for rollout labeling. This behavior measures oracle-task coverage; it does not provide observed-target matching or deployable target support. The separate source-audit path can match actor-visible descriptors to GT rows, but that audit does not change the generator's oracle-task source.

The primary evaluation direction is a fixed acquisition budget and bounded finite candidate support. First, an actor-visible myopic control must be admitted and calibrated on the same candidate table. Then bounded oracle lookahead establishes whether the candidate setup contains non-myopic headroom:

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

Only after meaningful preregistered headroom is present is the planned finite-H scorer evaluated for recovery:

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

Success is a matched held-out endpoint oracle evaluation, not a training loss or predicted value. The endpoint metric is:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

Its denominator is the root target error plus ε. The primary estimand is the equal-weight per-scene mean of paired endpoint differences, with the within-scene denominator equal to the number of paired, valid endpoint tasks in that scene and the scene-level denominator equal to the number of scenes with at least one such pair. The additive replay metric is the target-root gain, whose transition denominator is the maximum of the root target error and ε:

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

These are distinct metrics. The endpoint gain is an endpoint comparison with $(\Delta_0^e + \varepsilon)$, whereas additive target-root gain uses $\max(\Delta_0^e, \varepsilon)$; no shared-denominator or exact-equivalence claim is made. Its cumulative form is the additive trajectory diagnostic:

$$J_t^e = \sum_{k=0}^{t-1} r_k^e = \frac{\Delta_0^e - \Delta_t^e}{\max(\Delta_0^e, \varepsilon)}$$

State-relative RRI recomputes a state-specific denominator and remains a non-additive diagnostic. Secondary cumulative gain, state-relative RRI diagnostics, invalid-action rate, runtime, path length, and support coverage explain mechanism and feasibility.

No-root-action tasks, early-terminated trajectories, and no-supported-successor cases have no fixed-budget endpoint and are excluded from that endpoint denominator, never imputed as zero; each remains in the prespecified failure/

support strata, while any realized cumulative diagnostic is reported only through its last valid state. An oracle-evaluation failure likewise contributes no endpoint or label and is counted separately. These are missing endpoint observations and support evidence, not confirmatory policy results.

No confirmatory policy result is currently established. A missing target selector, unsupported endpoint artifact, insufficient candidate support, or mismatched source/profile contract blocks the downstream policy claim rather than becoming a zero result.

Gate	Evidence	Interpretation
Population	scene-split manifest and immutable counts	held-out inference population
Oracle task	GT task pool, sampled tasks, and failures	coverage evidence, not observed-target support
Myopic control	actor-visible ranking, calibration, and oracle re-evaluation	required control before planning
Planning headroom	bounded oracle lookahead versus one-step oracle greedy	finite-support opportunity gate
Finite-H policy	matched endpoint oracle artifact	planned recovery claim only after completion

Table 8: Objective-to-evidence gates.

5.2 Replay Eligibility and Finite-Horizon Learning Gate

5.2.1 Implemented evidence surface

The factual replay tables provide dense one-step labels only where a row is actor-selectable and has finite oracle support. `valid_action_mask`, `q_train_mask`, and padding remain distinct. Invalid and padded rows cannot enter selection, loss, or bootstrap. A nonterminal successor with no label-supported action has no bootstrap support; it is not assigned a utility value.

The implemented learner can consume selected transitions for an injected scorer. It does not establish a complete H=2 scorer, a task-sufficient successor memory, or a policy result. The primary planned learning objective is the bounded fixed-H return in additive target-root-gain units:

$$G_{t,e}^{(h)} = \sum_{k=0}^{\min(h,b_t)-1} \gamma^k r_{t+k}^e$$

For the admitted persisted Q__H primary path, the stored reward metric is `target_root_gain`; state-relative RRI is reported separately and is never substituted there. The generic `selected_target_reward` helper retains a legacy fallback through `target_root_gain`, `root_gain`, `target_rri`, or `rri` for non-admitted diagnostic rows outside Q__H evidence. That compatibility fallback is not the primary learning or evaluation path.

Surface	Current role	Minimum evidence
dense one-step	implemented label/read surface	actor-valid row and finite root-gain label
fixed-H scorer	planned primary objective	production scorer, compatible checkpoint, and supported selected transitions
endpoint evaluation	planned scientific proof	matched held-out oracle re-evaluation

Table 9: Learning surfaces and evidence gates.

The read-model evidence is bounded and factual. It aligns stored rows, preserves candidate identity, and reports padding and mask counts when explicitly requested; it does not manufacture support for unobserved transitions.

5.2.2 Development controls

Exact-$H = 2$ supervision is a base-case certification control. Requested-horizon recursion, Monte Carlo behavior returns, Double-Q comparisons, conservative offline controls such as CQL/BCQ, horizon-balanced sampling, and residual objectives are separate estimands or ablations. Each requires its own horizon/support report and must not be pooled into the primary fixed-H method.

Evaluation reports per-horizon admitted states, selected actions, support, terminal/no-bootstrap fractions, loss and ranking, while endpoint target-root gain remains the scientific outcome. No single aggregate loss can establish policy quality.

5.3 Policy Comparison and Statistical Protocol

The experimental unit is the scene. Comparisons are paired by scene, target task, root state, candidate-generation distribution, hard-validity regime, and fixed acquisition budget. Repeated snippets and seeds are aggregated within scene so densely sampled scenes do not dominate the endpoint estimand.

The primary comparison is the paired per-scene difference in fixed-budget endpoint target-root gain. The endpoint metric uses the canonical definition:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

Its denominator is $(\Delta_0^e + \varepsilon)$. The additive target-root-gain trajectory metric instead uses the canonical root-normalized reward and cumulative references, whose denominator is $\max(\Delta_0^e, \varepsilon)$. These metrics are distinct estimands; this protocol makes no shared-denominator or exact-equivalence claim. The analysis manifest must freeze the equal-weight scene aggregation: first average paired differences over complete endpoint pairs within each scene, then average those scene means over scenes with at least one pair. It must also freeze uncertainty, interval level, comparison family, exclusions, and minimum meaningful effect before confirmatory inspection. State-relative RRI is a per-transition diagnostic with a state-varying denominator; cumulative state-relative RRI is therefore not additive and is not the primary endpoint. Invalid-action rate, runtime, path length, and support coverage are also secondary diagnostics; none replaces the endpoint metric.

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

$$J_t^e = \sum_{k=0}^{t-1} r_k^e = \frac{\Delta_0^e - \Delta_t^e}{\max(\Delta_0^e, \varepsilon)}$$

The policy table keeps information boundaries explicit:

Policy	Information	Budget	Role
π_{rand}	actor-visible	H	valid-action reference
π_{myopic}	actor-visible	H	planned/calibrated one-step control
$\pi_{\text{oracle-1}}$	oracle immediate reward	H	one-step oracle comparator
$\pi_{\text{oracle-look}}$	bounded oracle lookahead	H	conditional finite-support upper reference
π_Q	actor-visible	H	planned fixed-H recovery

Table 10: Matched policy comparison with explicit information boundaries.

Oracle lookahead and one-step oracle greedy use one target mesh/crop, depth source, renderer/backprojection, fusion, point cap, and point-mesh metric contract. Repeated evaluation tests deterministic execution under that contract, not representation or metric independence. Learned policies use the same endpoint oracle and budget, but the learned comparison remains prospective until an actor-visible target-conditioned scorer and checkpoint exist.

Failure strata are retained: missing target task, oracle-evaluation failure, no valid root action, early termination, no supported successor, and completed paired trajectory. No valid root action means no fixed-budget endpoint is available; early termination and no supported successor likewise stop the trajectory at its last valid state and contribute no later gain. An oracle-evaluation failure invalidates the affected label or endpoint rather than creating a zero. These cases are excluded from the complete-pair endpoint denominator but remain in the declared support/failure population and are reported as such. A policy that cannot continue is not silently dropped, and incomplete artifacts do not establish headroom or superiority.

The interpretation is conditional on the frozen candidate support, actor/profile protocol, metric, and endpoint artifact. A failed recovery result can indicate target observability, action support, replay coverage, reward construction, state aliasing, or model capacity; these explanations must be separated by prespecified diagnostics rather than inferred from training loss.

6 Results

Validation TODO: Replace fixture- and readiness-oriented output with confirmatory results that answer each research question using matched endpoint evaluation, denominators, exclusions, aggregation units, independent-run uncertainty, and immutable artifact provenance.

Source: confirmatory report bundle and analysis manifest

Gate: submission evidence bundle passes all report assertions and every stated result resolves to raw and derived artifacts

The thesis report bundle is loaded through the strict schema checked in `experiment_data.typ`. Its declared evidence class is *pilot*. Schema validity establishes only that provenance, missingness, and table contracts are readable; it does not turn a development fixture or an incomplete pilot into scientific evidence.

Result	Required evidence	Status
Population and targets	population facts with denominators in every validated store	not available
Candidate support	valid/total candidates and zero-action failures in every validated store	not available
Paired policy effect	effect, interval, scene count, and headroom gate in every validated store	not available
Resource feasibility	wall time, peak GPU memory, and storage in every validated store	not available

Table 11: Evidence availability in the loaded report bundle. "Not available" means that no thesis result is inferred from fixture values, train-only attempts, or an incomplete store.

6.1 Candidate and Store Feasibility

The available artifacts show that the finite-candidate rollout path reaches mesh rendering, target-specific oracle scoring, and selected-action replay on training sources. They therefore support an implementation-readiness claim only. A CUDA out-of-memory failure in an unbatched candidate render and later memory-bounded attempts identify rendering as a scale gate; neither establishes rollout throughput, storage cost, candidate-family support, or policy quality for the intended study population.

6.2 Target-Task Coverage

The implemented pipeline samples geometry-valid GT target tasks and records target and validity fields in the rollout schema. The loaded bundle does not contain a validated population and exclusion table for the study. Scene coverage, eligible-target coverage, invalid-reason frequencies, and their denominators are consequently unavailable as results.

6.3 Policy Comparison

The primary estimand is not estimable from the current evidence because no validated held-out bundle contains the matched policy outcomes and aggregation inputs. Oracle repeatability, oracle-lookahead headroom, learned one-step performance, finite-horizon recovery, uncertainty intervals, and sensitivity analyses are therefore not reported.

6.4 Runtime and Storage Gate

The runtime and storage claim requires completed validated stores whose resolved manifests, failure records, and resource summaries refer to the same run. Until that evidence exists for both rollout profiles, the large-scale data-generation gate remains pending. The current attempts justify memory-bounded rendering and retained failure provenance, but no extrapolation to cluster throughput or dataset volume.

7 Discussion

Remove or rewrite before submission: Rewrite the discussion after confirmatory results exist. The present implementation-readiness narrative and architecture bridge registry are development material; the final chapter must interpret measured answers, alternative explanations, scope, limitations, and failure modes.

Source: Chapter 6 and the frozen analysis plan

Gate: every interpretation points to a reported result or is explicitly bounded as a limitation

The current evidence supports a narrow substrate conclusion. ARIA-NBV has an executable finite-candidate path that separates actor-visible policy inputs from oracle-only target geometry, applies invalidity as a hard decision constraint, evaluates target-specific reconstruction change offline, and exposes the resulting store through one typed reporting seam. This implementation is evidence toward the planned pc-c1-auditable-experiment-contract, but it does not establish pc-c1 or show that one candidate family, rollout policy, representation, or learned model performs better than another.

The rollout attempts also expose a systems limitation. Counterfactual scoring repeatedly renders a large mesh for a candidate set, so renderer memory and latency constrain feasible branch factor and rollout volume before statistical efficiency becomes relevant. An out-of-memory observation is evidence for batching and resource measurement, not for or against the candidate distribution or planning objective. Completed validated stores are required before throughput, storage, or failure rates are generalized.

The scientific limitations are more restrictive. Current attempts use training sources; no held-out paired policy table supports the endpoint estimand; metric repeatability and meaningful oracle-lookahead headroom remain unestablished; and no target-conditioned finite-horizon policy comparison has been demonstrated. The present evidence therefore cannot distinguish negligible non-myopic structure from inadequate candidate support, target observability, replay coverage, representation support, or model failure.

7.1 Evidence-conditioned architecture bridges

Implementation: exploratory **Evidence: pending**

Alternative representations and policies remain development hypotheses, but they are promoted only as responses to diagnosed bottlenecks.

Promotion gate: a measured limitation that the proposed bridge directly tests

Development source: Table 4; Table 7

If the local EFM3D field is the limiting variable, broad semidense memory, sparse occupied/free/unknown state, target-centred re-lifting, or logged appearance features provide progressively stronger representation tests. If compact descriptors lose relevant spatial structure, point, sparse, or equivariant encoders become justified. If independent candidate queries leave errors correlated with the valid candidate set, DeepSets or masked candidate interaction becomes relevant. If selected-view history fails specifically by approach direction, signed first- plus second-moment memory, followed only if needed by spherical harmonics, becomes the targeted ablation. A renderable 3DGS state is appropriate only when candidate rendering or soft instance membership is itself the measured missing capability.

Continuous and hierarchical policies change the action contract rather than merely increasing model capacity. A target-then-pose policy could first select or mask a target token and then condition pose queries on target, horizon, and step; multiple targets can share scene encodings and be evaluated in parallel. This remains post-core work because continuous pose generation requires separate feasibility, safety, support, and simulator-realism evidence. The finite-candidate model is retained as the calibrated control even if a continuous policy is later introduced.

Sequence or recurrent models are similarly conditional. They become scientifically motivated when errors persist after explicit history, time, horizon, and scene-memory conditioning, not simply because the dataset contains trajectories. Looping or recurrent refinement must preserve causal masks and candidate-row equivariance and must be compared against the same fixed-budget endpoint oracle evaluation.

Research TODO: After the primary evidence chain is complete, promote only bridges tied to a measured failure: EVL support, target observability, directional history, valid-set interaction, long-horizon credit, or action-support mismatch.

Source: method design-space registry

Gate: artifact-backed failure analysis

The next evidential step remains procedural: complete validated stores under resolved manifests, freeze the eligible population and exclusions, establish oracle stability and headroom, train the matched controls, and only then interpret architectural differences. The registry above preserves concrete follow-up hypotheses without presenting them as validated conclusions.

8 Conclusion

Validation TODO: Rewrite the conclusion as direct, evidence-calibrated answers to RQ1–RQ4. The current conditional outcome tree is useful development guidance but is not a final scientific conclusion.
Source: Chapter 6; Chapter 7
Gate: confirmatory results and discussion support one concise answer per research question

This thesis documents a leakage-auditable experiment substrate for target-conditioned finite-candidate next-best-view planning. The current implementation is evidence toward the planned pc-c1-auditable-experiment-contract: it covers the separation of actor-visible state from oracle supervision, the target-specific reconstruction objective, hard validity and replay contracts, and an artifact-driven reporting seam that keeps provenance and missingness attached to later results. It does not establish pc-c1; that contribution remains unreviewed and withheld until the downstream evidence gates are met.

The available evidence does not answer whether bounded oracle lookahead improves fixed-budget target reconstruction or whether a learned finite-horizon policy recovers such headroom. The current training-source rollout attempts provide evidence of pipeline reachability and reveal a renderer resource gate; the development report fixture documents the data contract only. Neither supports a held-out policy claim, a population-level effect, or a scale estimate.

The final scientific conclusion is therefore conditional on evidence that is not yet available. A stable oracle metric and positive paired lookahead effect would define measurable headroom for the evaluated finite support. Negligible headroom would be a setup-specific negative result. Unstable oracle evaluation would block planning claims, and stable headroom without learned recovery would remain a learned-control failure with several unresolved mechanisms. These outcomes delimit the thesis without extending it to continuous control, online reinforcement learning, or real-device deployment.

Development roadmap

This page records schedule, status, evidence pointers, and promotion gates. It is omitted from submission output. The active thesis owns claims; Python, tests, configuration, and immutable artifacts own behavior and measurements.

Implementation: partial **Evidence: pending**

Planning status only; this view promotes no policy-improvement claim.

Promotion gate: M1 correctness and M5 headroom evidence

Development source: `docs/typst/thesis/sections/`; `docs/typst/thesis/development/m1-contract-report.typ`; `aria_nbv/`; `aria_nbv/tests/`; active configuration

Outcome

The development outcome is a reproducible, evidence-backed thesis package. Canonical narrative and equation labels live in `docs/typst/thesis/sections/`; this page tracks readiness and gates only.

Milestones

- **M0 — scope and foundation (2026-04-29–2026-05-10): status: complete** gate: thesis scope, source policy, and citation owners aligned. Evidence: `docs/typst/thesis/main.typ`, `docs/typst/thesis/sections/01-research-questions.typ`, `docs/references.bib`.
- **M1 — data, cache, and oracle correctness (2026-05-11–2026-05-31): status: blocked** gate: fresh store, scene split, frame/depth, candidate alignment, Rerun, and throughput evidence. Evidence: `docs/typst/thesis/development/m1-contract-report.typ#ssec:m1-status`, `aria_nbv/tests/data_handling/`, `aria_nbv/tests/pose_generation/`, `aria_nbv/tests/rollouts/`, `.configs/offline_only.toml`.
- **M2 — VIN baseline and scale gate (2026-06-01–2026-06-21): status: blocked by M1** gate: reproducible baseline, diagnostics, and scale receipt. Evidence: `aria_nbv/tests/vin/`, `aria_nbv/tests/lightning/`, `docs/typst/thesis/sections/05-experimental-design/05-01-objectives-and-hypotheses.typ`, `docs/typst/`

thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ.

- **M3 — target-RRI and generation readiness (2026-06-22–2026-07-12): status: blocked by M1** gate: trusted target subset, actor-visible protocol, and deterministic generation checks. Evidence: `docs/typst/thesis/sections/03-oracle-and-data-generation/03-02-target-task-and-rri-labels.typ`, `aria_nbv/tests/oracle/test_target_selection.py`, `aria_nbv/tests/pose_generation/`, `aria_nbv/tests/rollouts/`.
- **M4 — target-conditioned one-step scorer (2026-07-13–2026-08-09): status: pending M3 evidence** gate: held-out ranking, calibration, oracle selections, and grouped failures. Evidence: `docs/typst/thesis/sections/04-method/04-02-descriptor-and-encoding-plan.typ`, `aria_nbv/tests/oracle/test_scoring.py`, `aria_nbv/tests/lightning/test_candidate_scorer_contract.py`.
- **M5 — lookahead headroom and Q_H (2026-08-10–2026-08-30): status: pending prerequisite gates** gate: equal-budget rollout comparisons, measured headroom, and supported Q_H evidence. Evidence: `docs/typst/thesis/sections/04-method/04-05-finite-candidate-value-model.typ`, `docs/typst/thesis/sections/05-experimental-design/05-02-learning-objective-and-replay-evidence.typ`, `aria_nbv/tests/data_handling/test_qh.py`, `aria_nbv/tests/rollouts/`.
- **M6 — scale and bridge decisions (2026-08-31–2026-09-13): status: pending M5** gate: scale decision and explicit bridge scope. Evidence: `docs/typst/thesis/sections/07-discussion.typ`, `aria_nbv/tests/oracle/test_online_vin.py`, `aria_nbv/tests/rollouts/test_reporting.py`.
- **M7 — final experiments and writing (2026-09-14–2026-09-27): status: pending M5/M6** gate: immutable run/config links, coverage statement, final tables, and failure cases. Evidence: `docs/typst/thesis/sections/06-results.typ`, `docs/typst/thesis/sections/07-discussion.typ`, `aria_nbv/tests/rollouts/test_reporting.py`.

- **M8 — release freeze (2026-09-28–2026-09-30): status: pending M7** gate: reproducible configs, docs, demo path, and smoke checks. Evidence: `Makefile`, `docs/README.md`, CI workflows, final evidence bundle.

Ablations

Comparison axes remain in canonical owners; this page records pointers and gates only.

- **Target input:** `docs/typst/thesis/sections/03-oracle-and-data-generation/03-02-target-task-and-rri-labels.typ`; gate: V0/V1 protocol tests and actor-visible boundary.
- **Candidate mixture:** `docs/typst/thesis/sections/03-oracle-and-data-generation/03-02-target-task-and-rri-labels.typ`, `aria_nbv/aria_nbv/pose_generation/`; gate: validity and provenance diagnostics.
- **Objective/planner:** `docs/typst/thesis/sections/05-experimental-design/05-01-objectives-and-hypotheses.typ`, `docs/typst/thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ`; gate: equal budget and oracle receipt.
- **Invalidity/learning:** `docs/typst/thesis/sections/04-method/04-04-architecture-contract.typ`, `docs/typst/thesis/sections/05-experimental-design/05-02-learning-objective-and-replay-evidence.typ`; gate: masks, reason codes, replay tests.
- **State/scale:** `docs/typst/thesis/sections/04-method/04-01-scene-representation-requirements.typ`, `docs/typst/thesis/sections/03-oracle-and-data-generation/03-03-replay-stores-and-diagnostics.typ`; gate: support and coverage accounting.

Evidence

- M1: `docs/typst/thesis/development/m1-contract-report.typ`, focused data/geometry/rollout tests, `.configs/offline_only.toml`, immutable store/Rerun receipts.

- M2–M4: `aria_nbv/tests/vin/`, `aria_nbv/tests/oracle/`, `aria_nbv/tests/lightning/`, and `docs/typst/thesis/sections/05-experimental-design/` plus saved calibration/ranking artifacts.
- M5–M7: rollout manifests, reports, configs, result tables, and exact support/coverage gaps in those artifacts.
- M8: CI, `make qmd-frontmatter-check`, `make thesis-pdf`, and final smoke receipts; tracked PDF remains a development artifact.

Risks

Active risk state is pointer-only: frame/depth `aria_nbv/tests/rendering/test_depth_backprojection_conventions.py`, target validity `aria_nbv/tests/oracle/test_target_selection.py`, rollout support `aria_nbv/tests/rollouts/test_zarr_store.py`. Open blockers remain in `docs/typst/thesis/development/m1-contract-report.typ` and `.agents/issues.toml`/`.agents/todos.toml`; no duplicate risk narrative is maintained here.

Freeze

Freeze gate: clean reproducible smoke matrix, immutable run/config links, final coverage statement, and no stale paths or placeholders. The exact submission evidence gate remains owned by `docs/typst/thesis/main.typ` and its report bundle; this development projection does not satisfy it.

Promotion queue

Promotion queue — candidate: Promote M5 lookahead/Q_H results after paired held-out evidence is immutable.
Source: docs/typst/thesis/sections/05-experimental-design/; aria_nbv/tests/rollouts/test_reporting.py target: docs/typst/thesis/sections/06-results.typ and docs/typst/thesis/sections/07-discussion.typ gate: positive headroom, oracle re-scoring, uncertainty, and coverage

Promotion queue — blocked: Resolve the scale fallback before claiming population coverage.
Source: docs/typst/thesis/development/m1-contract-report.typ; aria_nbv/tests/data_handling/; aria_nbv/tests/rollouts/ target: docs/typst/thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ gate: throughput and scene-level held-out coverage

Promotion queue — deferred: Keep online discrete Q__H and continuous target-then-pose work as bridge scope until offline gates close.

Source: docs/typst/thesis/sections/07-discussion.typ; .agents/issues.toml target: docs/typst/thesis/sections/07-discussion.typ gate: stable offline Q__H and explicit scope decision

Promotion queue — rejected: Do not promote unsupported continuous, VLM, semantic-global, or real-device claims into the thesis core.

Source: docs/typst/thesis/main.typ; docs/typst/thesis/sections/07-discussion.typ target: docs/typst/thesis/sections/08-conclusion.typ gate: bounded non-claim retained

Priorities

Gate order is M1 correctness, M2–M4 target/scorer readiness, M5 headroom, M6 bridge decision, then M7/M8 freeze. Each transition requires the evidence pointer and exit gate above; no scale or bridge work bypasses a blocked prerequisite.

Issues and blockers

- M1 store, scene split, Rerun, and throughput: `docs/typst/thesis/development/m1-contract-report.typ#m1-blockers`.
- M5 support and headroom: `docs/typst/thesis/sections/05-experimental-design/05-02-learning-objective-and-replay-evidence.typ`, `docs/typst/thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ`.
- Actionable follow-up: `.agents/issues.toml` and `.agents/todos.toml`; these records do not replace canonical owners.

M1 Contract Report

Status

Implementation: partial **Evidence: pending**

Synthetic contract guards are available, but the configured local store and scale evidence do not yet close the M1 exit gate.

Promotion gate: fresh store, scene-split, Rerun, and throughput evidence

Development source: thesis M1 gate notes and executable package owners (not a second contract owner)

Area	Status	Owner and evidence pointer
Source-contract guards	validated	Focused tests under `aria_nbv/tests/data_handling/`, `aria_nbv/tests/pose_generation/`, and `aria_nbv/tests/rollouts/` own the executable checks; this view records only their M1 status.
Configured local store	blocked	`.configs/offline_only.toml` and the resolved `.data/offline_cache/vin_offline` artifact are the smoke-evidence owners.
Scene-level split evidence	blocked	The store artifact and its split metadata require a scene-level train/validation proof before promotion.
Rerun diagnostics	blocked	No M1 Rerun recordings are present in this worktree. The recording gate remains open; absence is reported explicitly rather than linked to a nonexistent artifact.
Oracle throughput	blocked	Representative mesh, rendering, backprojection, and RRI timing remains to be captured in the experiment evidence owner.

Evidence

The current evidence view is intentionally pointer-only. The defining Python modules, tests, active configuration, and generated artifacts remain authoritative;

no schema, tensor shape, formula, serialized value, or command ledger is reproduced here.

- Synthetic guard results: `aria_nbv/tests/data_handling/`, `aria_nbv/tests/pose_generation/`, and `aria_nbv/tests/rollouts/`.
- Local smoke configuration and artifact root: `.configs/offline_only.toml` and `.data/offline_cache/vin_offline/`.
- Rerun evidence: absent in the current worktree; the normal/boundary/failure recording gate remains open.

Blockers

Promotion queue — blocked: Refresh the configured-store evidence and report coverage status.

Source: M1 gate notes and executable package owners target: M1 status and evidence gate: fresh immutable-store inspection

Promotion queue — blocked: Replace the one-scene diagnostic split with scene-level evidence or record the exact residual blocker.

Source: `.data/offline_cache/vin_offline/` target: M1 evidence gate: scene-level split proof

Promotion queue — blocked: Collect normal, boundary, and failure Rerun recordings.

Source: `.artifacts/rerun/` target: M1 evidence gate: three diagnostic recordings

Promotion queue — blocked: Measure representative oracle throughput and preserve the measurement artifact for review.

Source: `aria_nbv/` and active experiment configuration target: M1 evidence gate: representative timing receipt

Bibliography

[Baj88] R. Bajcsy, “Active Perception,” *Proceedings of the IEEE*, vol. 76, no. 8, pp. 966–1005, 1988, doi: 10.1109/5.5968.

[AWB88] J. Aloimonos, I. Weiss, and A. Bandyopadhyay, “Active Vision,” *International Journal of Computer Vision*, vol. 1, no. 4, pp. 333–356, 1988, doi: 10.1007/BF00133571.

[SRR03] W. R. Scott, G. Roth, and J.-F. Rivest, “View Planning for Automated Three-Dimensional Object Reconstruction and Inspection,” *ACM Computing Surveys*, vol. 35, no. 1, pp. 64–96, 2003, doi: 10.1145/641865.641868.

[Fra+25] N. Frahm *et al.*, “VIN-NBV: A View Introspection Network for Next-Best-View Selection.” [Online]. Available: https://arxiv.org/abs/2505.06219

[Eng+23] J. Engel *et al.*, “Project Aria: A New Tool for Egocentric Multi-Modal AI Research.” [Online]. Available: https://arxiv.org/abs/2308.13561

[Met25] Meta Platforms Inc., “Aria Synthetic Environments Dataset.” [Online]. Available: https://facebookresearch.github.io/projectaria_tools/docs/open_datasets/aria_synthetic_environments_dataset

[Str+24] J. Straub, D. DeTone, T. Shen, N. Yang, C. Sweeney, and R. Newcombe, “EFM3D: A Benchmark for Measuring Progress Towards 3D Egocentric Foundation Models.” [Online]. Available: https://arxiv.org/abs/2406.10224

[Met25] Meta Platforms Inc., “Egocentric Voxel Lifting (EVL) documentation.” [Online]. Available: https://facebookresearch.github.io/projectaria_tools/docs/open_models/evl

[Jia+25] Z. Jia, Y. Li, Q. Hao, and S. Zhang, “PB-NBV: Efficient Projection-Based Next-Best-View Planning Framework for Reconstruction of Unknown Objects,” *IEEE Robotics and Automation Letters*, vol. 10, no. 7, pp. 7444–7451, 2025, doi: 10.1109/LRA.2025.3573631.

[Kum+20] A. Kumar, A. Zhou, G. Tucker, and S. Levine, “Conservative Q-Learning for Offline Reinforcement Learning,” in *Advances in Neural Information Processing Systems*, 2020, pp. 1179–1191. [Online]. Available: https://papers.nips.cc/paper/2020/hash/0d2b2061826a5df3221116a5085a6052-Abstract.html

[FMP19] S. Fujimoto, D. Meger, and D. Precup, “Off-Policy Deep Reinforcement Learning without Exploration,” in *Proceedings of the 36th International Conference on Machine Learning*, in Proceedings of Machine Learning Research, vol. 97. PMLR, 2019, pp. 2052–2062. [Online]. Available: https://proceedings.mlr.press/v97/fujimoto19a.html

[HGS15] H. van Hasselt, A. Guez, and D. Silver, “Deep Reinforcement Learning with Double Q-learning.” [Online]. Available: https://arxiv.org/abs/1509.06461

[Zah+17] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Poczos, R. Salakhutdinov, and A. J. Smola, “Deep Sets,” in *Advances in Neural Information Processing Systems*, 2017. [Online]. Available: https://papers.nips.cc/paper_files/paper/2017/hash/f22e4747da1aa27e363d86d40ff442fe-Abstract.html

[Lee+19] J. Lee, Y. Lee, J. Kim, A. R. Kosiorek, S. Choi, and Y. W. Teh, “Set Transformer: A Framework for Attention-based Permutation-Invariant Neural Networks,” in *Proceedings of the 36th International Conference on Machine Learning*, in Proceedings of Machine Learning Research, vol. 97. PMLR, 2019, pp. 3744–3753. [Online]. Available: https://proceedings.mlr.press/v97/lee19d.html

[Wu+24] X. Wu *et al.*, “Point Transformer V3: Simpler, Faster, Stronger,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. [Online]. Available: https://arxiv.org/abs/2312.10035

[CGS19] C. Choy, J. Gwak, and S. Savarese, “4D Spatio-Temporal ConvNets: Minkowski Convolutional Neural Networks,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019. [Online]. Available: https://arxiv.org/abs/1904.08755

[SHW21] V. G. Satorras, E. Hoogeboom, and M. Welling, “E(n) Equivariant Graph Neural Networks,” in *Proceedings of the 38th International Conference on Machine Learning*, in Proceedings of Machine Learning Research, vol. 139. PMLR, 2021, pp. 9323–9332. [Online]. Available: https://proceedings.mlr.press/v139/satorras21a.html

[Bro+21] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković, “Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges.” [Online]. Available: https://arxiv.org/abs/2104.13478

[Zho+19] Y. Zhou, C. Barnes, J. Lu, J. Yang, and H. Li, “On the Continuity of Rotation Representations in Neural Networks,” *arXiv preprint arXiv:1812.07035*, 2019, [Online]. Available: https://arxiv.org/abs/1812.07035

[Zho+23] Z. Zhou, J. Wang, Y.-H. Li, and Y.-K. Huang, “Query-Centric Trajectory Prediction,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[e3n25] e3nn contributors, “e3nn spherical harmonics documentation.” [Online]. Available: https://docs.e3nn.org/en/stable/api/o3/o3_sh.html

[GML22] A. Guédon, P. Monasse, and V. Lepetit, “SCONE: Surface Coverage Optimization in Unknown Environments by Volumetric Integration,” in *Advances in Neural Information Processing Systems*, 2022. [Online]. Available: https://arxiv.org/abs/2208.10449

[JLD24] W. Jiang, B. Lei, and K. Daniilidis, “FisherRF: Active View Selection and Uncertainty Quantification for Radiance Fields using Fisher Information.” [Online]. Available: https://arxiv.org/abs/2311.17874

[GE19] Y. Geifman and R. El-Yaniv, “SelectiveNet: A Deep Neural Network with an Integrated Reject Option,” in *Proceedings of the 36th Inter-*

national Conference on Machine Learning (ICML), in Proceedings of Machine Learning Research, vol. 97. PMLR, 2019, pp. 2151–2159. [Online]. Available: https://proceedings.mlr.press/v97/geifman19a.html

[Liu+19] Z. Liu, Z. Wang, P. P. Liang, R. Salakhutdinov, L.-P. Morency, and M. Ueda, "Deep Gamblers: Learning to Abstain with Portfolio Theory," in *Advances in Neural Information Processing Systems*, 2019, pp. 10622–10632. [Online]. Available: https://proceedings.neurips.cc/paper/2019/hash/0c4b1eeb45c90b52bfb9d07943d855ab-Abstract.html

[Ben+09] Y. Bengio, J. Louradour, R. Collobert, and J. Weston, "Curriculum Learning," in *Proceedings of the 26th International Conference on Machine Learning (ICML)*, 2009. [Online]. Available: https://proceedings.mlr.press/v5/bengio09a.html

[Met25] Meta Platforms Inc., "PyTorch3D Cameras + Rendering Documentation, Tutorial: Render a Textured Mesh." [Online]. Available: https://pytorch3d.readthedocs.io/en/latest/modules/renderer/cameras.html,%20https://pytorch3d.org/docs/renderer,%20https://pytorch3d.org/docs/renderer_getting_started,%20https://pytorch3d.org/docs/cameras,%20https://pytorch3d.org/tutorials/render_textured_meshes

[DAT20] N. De Cao, W. Aziz, and I. Titov, "The Power Spherical distribution." [Online]. Available: https://arxiv.org/abs/2006.04437

A Reproducibility Record

The report bundle is the single numerical interface between validated rollout artifacts and this manuscript. Its schema version, evidence status, store manifests, typed parameter rows, failure records, and sidecar hashes preserve provenance without duplicating analysis logic in Typst. The document loader rejects schema or status mismatches before any table is rendered. Compact labels and digest prefixes are used below; complete store identifiers, paths, and hashes remain in the machine-readable bundle.

Evidence class	Store	Manifest digest	Validated
Development fixture	store S1	not-a-real-m...	yes

Table 12: Loaded development-fixture provenance. The fixture verifies the document interface only and is not scientific evidence.

The bundled development fixture contains synthetic values solely to test typed loading and missingness. Its numerical rows are deliberately not rendered or interpreted.

B Development Diary

Archived source note: This marked diary is a development-only record. It is excluded from submission mode; only evidence-backed definitions, protocols, and results graduate into the thesis body.

Source: thesis mode contract

B.1 Decisions Retained

The thesis keeps the privileged oracle and actor-visible policy as separate contracts, evaluates finite candidate sets under a shared validity mask, and treats target-specific reconstruction improvement as the common comparison signal. Rollout stores retain selected transitions and provenance rather than becoming a second implementation of the oracle or training pipeline.

Open decision: Freeze the scene-disjoint analysis population and the minimum evidence scale only when the resolved manifests and exclusion ledger are available.

Source: experimental-design contract

Gate: confirmatory bundle freeze

B.2 Failures That Changed the Plan

Early rollout attempts reached the mesh-rendering path but exceeded available accelerator memory when candidate views were rendered without a bounded batch. This failure narrowed the immediate claim to implementation readiness and made peak memory, throughput, failure provenance, and completed-store validation explicit scale gates.

Validation TODO: Require completed stores, matching manifests, resource summaries, and recorded failures before extrapolating rollout bandwidth or storage demand.

Source: rollout attempt artifacts

Gate: cluster-readiness evidence

B.3 Rejected Scope

Continuous actions, online simulators, additional scene representations, and alternative reinforcement-learning families are excluded from the core study while the finite-candidate target-conditioned comparison remains unvalidated. They would change both the action contract and the supervision regime, so adding them now would weaken rather than extend the primary experiment.

Remove or rewrite before submission: Delete bridge material that cannot be tied to a concrete limitation or a matched follow-up experiment after the confirmatory analysis is complete.

Source: scope review

Gate: discussion freeze

B.4 Directional-Memory Hypothesis

Research TODO: Treat target-centred directional memory as an ablation hypothesis, not an actor feature or contribution. The representation must be implemented, mask-audited, and compared against pose distance and overlap before any predictive claim enters the manuscript.

Source: RQ3 representation hypothesis

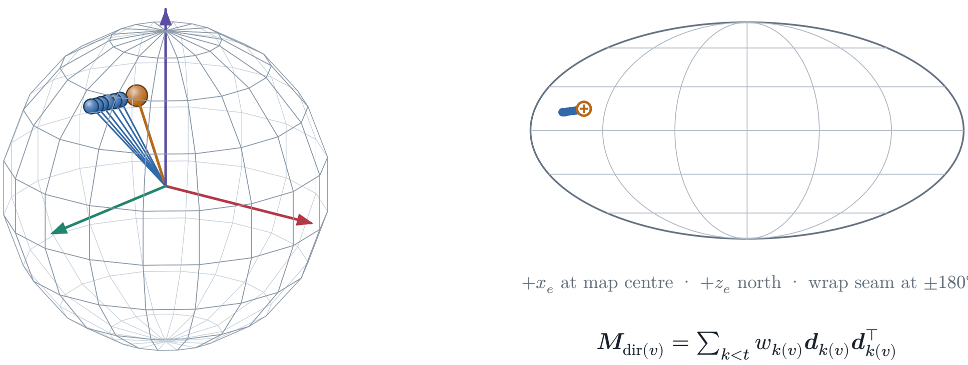
Gate: paired held-out ablation

A. Target-centred directions on $\mathbb{S}^2$

actual logged camera centres, expressed in the target-object frame

B. Complete-domain view and prospective compression

Mollweide is equal-area; no smooth field or SH lobe is implied



● logged history direction ⊕ example query

Scene 81286, sample ASE_81286_Atek_000024, target instance 128 frames 0, 2, 4, 6, 8, 10 and query frame 15.

$$M_{\mathrm{dir}(v)} = \textstyle\sum_{k<t} w_{k(v)} \boldsymbol{d}_{k(v)} \boldsymbol{d}_{k(v)}^\top$$

$$\nu_{i(v)} = 1 - \frac{\boldsymbol{d}_i^\top \boldsymbol{M}_{\mathrm{dir}} \boldsymbol{d}_i}{\operatorname{tr} \boldsymbol{M}_{\mathrm{dir}} + \varepsilon}$$

This fixture uses uniform weights and a logged future pose only to make the second-moment query inspectable. It does not claim that the learner currently stores $\boldsymbol{M}_{\mathrm{dir}}$ or that this novelty predicts target RRI.

HYPOTHESIS / ABLATION MATERIAL. Keep outside submission claims until the representation exists and paired evaluation tests whether it adds information beyond pose distance and overlap.

Figure 7: Development-only directional-memory fixture. The sphere and Mollweide panels show the same logged ASE camera directions in a target-object frame; the orange query is another logged pose used to inspect one prospective second-moment novelty definition. No smooth field, spherical-harmonic representation, or learned benefit is claimed.

B.5 Next Evidence Gates

The next admissible evidence is a validated report bundle that fixes the study population, records target and candidate exclusions, supplies paired endpoint outcomes with uncertainty, and joins runtime and storage statistics to the same manifests. Only then can the manuscript replace the present availability statements with policy comparisons.

Implementation TODO: Export the confirmatory report bundle, compile both thesis modes against it, and inspect the resulting tables and failure cases before freezing claims.

Source: reporting contract

Gate: claim freeze

Ehrenwoertliche Erklaerung

Ich versichere, dass ich diese Masterarbeit selbststaendig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Muenchen, 7. Jan 2027 Jan Duchscherer